



Review article

Monetizing battery electric vehicle flexibility in Europe: A meta-analysis on the economic value of intelligent on-site and off-site energy system integration

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ABSTRACT

Battery electric vehicles are increasingly recognized not only as zero-emission transport technologies but also as flexible energy assets that can support modern power systems. While numerous studies have demonstrated the technical feasibility of utilizing vehicle-based flexibility, its economic potential remains difficult to assess due to heterogeneous evaluation frameworks, inconsistent assumptions, and non-comparable reporting metrics across studies.

To address this fragmentation, this review conducts a structured meta-analysis of 25 peer-reviewed publications, extracting 165 individual scenario results, each representing a scenario-specific economic outcome of an electric vehicle fleet in a defined use case, including underlying assumptions on fleet configuration, market conditions, and operational strategy. A standardized evaluation framework is introduced that expresses economic value relative to the underlying flexible energy and power capacities, enabling consistent cross-study comparison.

The analysis covers a broad range of on-site (behind-the-meter) and off-site (front-of-the-meter) use cases, including self-consumption optimization, peak shaving, tariff-optimized charging, spot market trading, and ancillary services, with a focus on European energy systems. The results indicate that off-site applications, particularly intraday trading and participation in automatic frequency restoration reserve, tend to offer the highest monetization potential, exceeding 20k/(MWh) and 100k/(MW) under bidirectional integration. In contrast, on-site strategies provide lower but more stable economic benefits with reduced market complexity.

By introducing physically grounded and transferable metrics, this study enhances comparability across existing literature and provides a transparent basis for evaluating the economic value of battery electric vehicle flexibility in different application contexts.

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1. Introduction

The accelerating electrification of passenger and commercial road transport, driven by climate goals and rapid technological progress, is fundamentally reshaping the intersection of mobility and energy systems (Li, 2024). Battery electric vehicles (BEVs), now emerging as flexible, distributed energy resources, introduce both significant opportunities and complex challenges for power systems already in transition, shaped by the rise of renewable energy (Mladenov et al., 2021), increasing industrial electrification (Wei et al., 2019), and evolving market dynamics (Thomaßen et al., 2022).

Beyond their primary mobility tasks, BEVs may unlock an additional economic potential due to their flexible battery storage capacities during idle periods. Aggregated fleet battery capacity can be leveraged to optimize local energy consumption patterns, support public grid operation, and participate in electricity markets, creating value streams that internal combustion engine vehicles cannot offer (Lei et al., 2025). Many stakeholders are increasingly focused on assessing the emerging economic potential of BEV fleets in these secondary applications. Original Equipment Manufacturers (OEMs) are adapting product strategies; fleet operators are evaluating investments in charging infrastructure and energy management; financial institutions (such as investors, insurance companies and leasing firms) are refining pricing and risk models; and market aggregators and grid operators are exploring new mechanisms to harness the flexibility of BEV fleets, for example through virtual power plants (VPPs) and flexibility markets (Kaiss et al., 2025; Ostermann and Koetter, 2016; Möbius et al., 2023; Al-Hanahi et al., 2021).

This potential is unlocked by two modes of energy system integration: unidirectional and bidirectional (Dioha et al., 2022). Advanced unidirectional integration typically involves smart charging, optimizing charging schedules to coincide with low electricity prices or periods of reduced grid demand, minimizing energy costs, and alleviating system stress if considered in the optimization. Bidirectional approaches, such as vehicle-to-home (V2H) or vehicle-to-site (V2S), and vehicle-to-grid (V2G), allow vehicles to discharge electrical energy back into local systems or the grid. While these bidirectional applications can unlock substantial economic and system-level benefits, they introduce increased investment cost and additional battery cycling, potentially accelerating degradation and shortening battery life. This necessitates a key trade-off between economic investment, operational value generation, and asset longevity (Sagaría et al., 2025; Bhoir et al., 2021; Menchaca Santos et al., 2025).

A growing body of research has sought to operationalize the economic potential of BEV integration by developing advanced algorithmic solutions. These studies propose diverse approaches for optimizing the participation of BEVs in both local energy systems (on-site) and broader electricity markets (off-site) (see Fig. 1). On-site integration mainly focuses on energy management strategies that use BEVs to optimize residential or commercial energy consumption patterns, enhance self-consumption of on-site generated renewable energy (typically photovoltaic (PV)), and improve resilience during grid outages. Algorithmic solutions in this domain often involve multi-objective optimization to balance competing goals such as minimizing electricity costs, reducing peak demand, and maintaining adequate state-of-charge (SOC) for future mobility needs. Common approaches include model predictive control (MPC) frameworks that dynamically schedule charging and discharging based on forecasts of household or site demand, solar generation, and electricity prices (Möbius et al., 2023; Rosner et al., 2025; Yoshimura et al., 2015; Yousefi et al., 2021; Ohno et al., 2024; Blonsky et al., 2022), as well as rule-based heuristics that prioritize simplicity and low computational overhead (Wang et al., 2024; Chen et al., 2014; Aznavi et al., 2020). Some studies explore real-time learning-based control schemes for responding to unexpected changes in load or PV output (Huy et al., 2023; Amer et al., 2023; Almughram et al., 2023; Forootani et al., 2022; Aldahmashi and Ma, 2024).

In parallel, off-site strategies allow BEVs to serve the public grid, unlocking additional value streams. Trading algorithms maximize revenues from energy arbitrage in day-ahead markets (Xu et al., 2017; Zheng et al., 2020; Aliasghari et al., 2020; Kern et al., 2020), intraday markets (Kern et al., 2020; Naharudinsyah and Limmer, 2018; Tepe et al., 2022; Teske et al., 2021; Chemudupaty et al., 2025), and across multiple coordinated energy markets (Kern et al., 2020; Vollmuth et al., 2024). Other strategies enable BEVs to provide ancillary services, such as frequency regulation and spinning reserves (Tepe et al., 2022; Seo et al., 2024; Will and Ocker, 2024; Andersen et al., 2018; Figgenger et al., 2022; Lunde et al., 2024). Meaningful participation in these markets is supported by numerous studies on aggregating individual BEVs into VPPs or fleet-level control architectures capable of delivering reliable, dispatchable flexibility (Pertl et al., 2019; Zhou et al., 2021; Brinkel et al., 2023). These approaches often rely on hierarchical or distributed control schemes coordinating charging and discharging decisions across many vehicles while accounting for user mobility requirements and infrastructure constraints. Complementing these control strategies, other studies have emphasized the importance of accurate forecasting of BEV flexibility potential, including

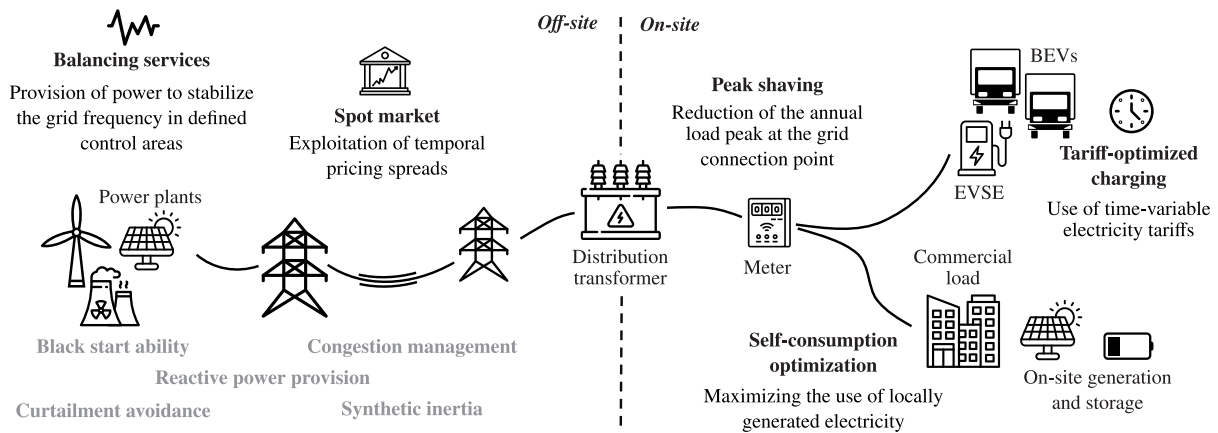


Fig. 1. Established and upcoming secondary use cases to reduce operating expenses for (non-) commercial battery electric vehicle fleets. Revenue-generating use cases beyond the grid connection point, on the grid side, are classified as off-site strategies, while those generating revenue by leveraging intelligent energy management on the depot side are categorized as on-site strategies. Use cases covered by this meta-analysis are highlighted in black, while other emerging opportunities not covered in the analysis are shown in gray.

predictions of plug-in availability, SOC trajectories, and mobility patterns (Pertl et al., 2019; Brinkel et al., 2023; Visser et al., 2022). Robust forecasting models are essential to reduce market participation uncertainty and ensure the reliable delivery of committed services. Together, these contributions form a growing foundation for integrating BEV fleets into real-time grid operations and energy markets, positioning them as a scalable source of system flexibility.

While the technical feasibility of integrating BEVs into energy systems has been widely explored, a growing subset of the literature has shifted focus toward quantifying the economic potential of these strategies. As stakeholders seek to justify investments in charging infrastructure, energy management systems, and market participation mechanisms, understanding the potential financial returns from intelligent BEV integration becomes essential. Although numerous studies have examined the economic benefits of intelligent on- and off-site integration and underlying secondary use cases, the overall picture remains unclear. Comparisons are challenging, as existing studies employ heterogeneous evaluation metrics (e.g., per-vehicle revenues, total fleet value, or case-specific cost savings), differing temporal and spatial system boundaries, and inconsistent assumptions regarding market conditions, fleet characteristics, and operational strategies. As a result, reported economic outcomes are often not directly comparable, limiting their interpretability and transferability across applications and studies, particularly when differences in aggregation level, usable flexibility, and market context are not explicitly accounted for.

Beyond differences in reporting formats, key modeling assumptions used to quantify economic outcomes vary substantially across studies. For instance, some analyses report results on a per-vehicle basis, while others consider aggregated fleet-level values. Furthermore, the usable flexibility is defined inconsistently, with some studies assuming access to the full battery capacity, whereas others restrict operation to a limited SOC window (e.g., 30%–70%) to account for mobility constraints or battery degradation. Inconsistencies also arise from differing market contexts and technical configurations. Studies focus on different geographic markets, which exhibit distinct price dynamics, regulatory conditions, and remuneration schemes.¹ At the same time,

¹ This study focuses explicitly on European energy systems. This regional scope is motivated by the relatively high maturity and heterogeneity of European electricity markets, including well-established spot and balancing markets, advanced regulatory frameworks for flexibility participation, and increasing integration of renewable energy sources. These characteristics make Europe a particularly relevant and data-rich environment for assessing the economic potential of BEV flexibility. The proposed methodology itself is transferable and can be applied to other regions in future work.

analyzed vehicle fleets vary significantly in battery capacity, (dis-)charging power, and operational patterns. These differences directly influence the monetization potential and further complicate cross-study comparability.

In response to these challenges, several systematic reviews and meta-analyses have been conducted to synthesize revenue estimates across different use cases and modeling frameworks. However, despite these efforts, a key methodological limitation remains: existing reviews and meta-analyses primarily aggregate results without establishing a consistent, physically grounded basis for comparison across studies. In this context, a structured meta-analysis is particularly suitable, as it enables the systematic extraction, normalization, and comparison of quantitative results across heterogeneous studies with differing assumptions, system boundaries, and reporting formats. By transforming scenario-specific outcomes into standardized metrics relative to underlying energy and power flexibility capacities, such an approach addresses the lack of comparability in the literature and enables a more consistent and transferable assessment of economic potential.

This limitation becomes apparent when examining how economic outcomes are reported and compared in the existing literature. For instance, Signer et al. (2023) present a systematic literature review evaluating the economic potential of V2G in spot and balancing markets. Based on an analysis of 81 publications, they identify a wide range of annual revenue estimates per BEV, reflecting the diversity of modeling approaches, such as deterministic and stochastic optimization, cost-benefit analysis, and simulation, as well as variation across market types and global regions. Complementing this perspective, Heilmann and Friedl (Heilmann and Friedl, 2021) conduct a meta-analysis of 340 cases from 35 studies to identify the key factors influencing the economic viability of unidirectional and bidirectional charging applications in energy systems. Their analysis finds that grid service applications yield the highest revenue potential. Furthermore, the study highlights that technical parameters such as charging power and efficiency, battery capacity, and bidirectional charging capability are critical in enhancing economic performance. Adding to this body of work, Thompson and Perez (Thompson and Perez, 2020) conduct a meta-analysis of 15 studies to evaluate the value stream potential of vehicle-to-anything (V2X) integration across markets in Australia, Canada, the US, and the UK. They introduce a V2X value stream framework distinguishing between energy-based and power-based services, showing that power-based services typically cause less battery degradation and enable greater stacking opportunities. The analysis further underscores the importance of regulatory environments in shaping the economic viability of V2X applications. Similarly, Ghatikar and

Alam (Ghatikar and Alam, 2023) review 11 demonstration projects in the US and Europe to explore the techno-economic implications of V2H and V2G applications. Their findings highlight revenue opportunities and strategic challenges for OEMs, emphasizing the need for innovation in business models to realize continuous value from bidirectional services. In addition to these synthesis-oriented studies, a broader body of literature, including case studies, simulation-based analyses, and regulatory reports, has examined the economic valuation of BEV flexibility from diverse perspectives, often focusing on specific use cases, market designs, or system configurations.

While previous meta-analyses have made important strides in evaluating the economic potential of BEV fleet integration into energy systems, a key limitation in cross-study comparability remains. This is largely due to the absence of a consistent, physically grounded basis for evaluating economic outcomes relative to the underlying flexibility potential. To address this gap, this work adopts a structured meta-analytical approach to systematically extract and harmonize quantitative results from the literature. By introducing standardized metrics that relate economic outcomes to underlying energy and power flexibility capacities, the study establishes a consistent basis for cross-study comparison. This enables a more transparent, scalable, and transferable assessment of the economic potential of BEV flexibility across a wide range of on-site and off-site secondary use cases.

1.1. Contributions

The key innovation lies in systematically quantifying the economic potential of BEV fleet flexibility across various secondary use cases. Benchmarking revenues against aggregated idle energy and power capacities establishes a consistent and scalable basis for comparison across studies. The results enhance transparency and transferability and support informed, data-driven decision-making for researchers, industry practitioners, and policymakers. In this context, the main contributions of this study are:

- **Introduction of standardized normalization metric** that express the monetization potential of BEV fleet flexibility relative to underlying energy and power capacities. This enables consistent, per-unit economic comparisons across studies with diverse modeling approaches, vehicle types, and market contexts.
- **Meta-analysis of 165 data points quantifying the economic potential of BEV fleet flexibility** in Europe. Each data point represents a scenario comparing smart charging against an uncoordinated baseline, covering a wide range of secondary use cases. On the one hand, these include on-site applications such as self-consumption optimization, tariff-optimized charging, and peak shaving. On the other hand off-site applications like trading on the day-ahead spot market, intraday auction, and continuous spot markets, and participation in ancillary services including frequency containment reserve and automatic frequency restoration reserve. Both unidirectional and bidirectional integration strategies are considered.

1.2. Organization of the article

Section 2 provides background information on definitions of the secondary on-site and off-site use cases examined in this study. Section 3 outlines the methodological approach and describes the selection criteria for analyzed articles. The results, indicating the economic potential of aggregated electric vehicle batteries in secondary use cases, compensated for both energy and power, are presented in Section 4. Finally, Section 5 summarizes the key findings and offers an outlook on future research directions.

2. Secondary use cases

In this work, on-site and off-site use cases are distinguished based on their system boundary relative to the point of grid connection. On-site use cases refer to strategies implemented behind the meter, where flexibility is utilized to optimize local energy consumption and reduce electricity costs. In contrast, off-site use cases involve the provision of flexibility to external actors, markets, or grid services outside the local energy management objective, thereby enabling direct revenue streams.

This spatial classification is orthogonal to the commonly used distinction between implicit and explicit flexibility. While implicit flexibility typically refers to price-based responses (e.g., time-varying tariffs) and explicit flexibility to direct market participation or contractual service provision (Bovera and Rancilio, 2024), both forms can occur within the on-site and off-site use cases considered in this work. For example, tariff-optimized charging represents an on-site strategy following external (implicit) price signals, whereas participation in balancing markets constitutes an off-site (explicit) flexibility provision.

Section 2.1 provides a brief definition and description of secondary use cases on-site. Subsequently, Section 2.2 provides an overview of revenue-generating use cases beyond the grid connection point, off-site. A schematic representation of on-site and off-site integration is shown in Fig. 1. Furthermore, Table 1 lists key market characteristics relevant to the secondary use cases examined in this work.

2.1. On-site

On-site energy use cases refer to energy management strategies and technologies implemented behind the utility meter, directly at the customer's premises, such as depots or fleet charging locations. These use cases support demand-side management (DSM) and focus on reducing operational costs by optimizing energy consumption patterns, increasing locally generated renewable electricity use, and responding intelligently to dynamic pricing signals. As the savings are realized on the customer side of the meter, these approaches do not generate external revenue, but instead lower electricity expenses, infrastructure costs, or both, thereby enhancing the site's economic viability. The boundaries between different on-site use cases are not always clear-cut, as they tend to overlap in practical operation.

2.1.1. Peak shaving

Peak shaving (PS), also referred to as load shedding or peak load shaving, is an on-site use case designed to reduce a facility's maximum electricity demand, thereby lowering associated electricity expenses. This approach aims to flatten the demand curve and avoid demand spikes by redistributing electrical loads to off-peak periods. Peak shaving (PS) focuses on controlling energy consumption on the customer side to stabilize the grid and reduce costs (Zhong et al., 2010; Borlaug et al., 2021). The economic benefits of reducing peak load manifest in several ways. First, network charges, which comprise a substantial portion of electricity costs, are typically based on recorded highest demand. Reducing peak consumption directly lowers these charges (Jankowiak et al., 2020; Bayernwerk, 2025; Ostermann et al., 2022). Second, lower peak demand can reduce the need for extensive physical grid infrastructure, such as large transformer stations and power lines, since the cost of installing, operating, and maintaining these assets scales with the maximum power requirement (Borlaug et al., 2021). Finally, some electricity providers offer more favorable tariffs to consumers with lower or more predictable peak loads, allowing for additional cost optimization (Hj Osman et al., 2024).

BEV fleet operators can support PS by strategically shifting charging activities to off-peak hours, thus avoiding load spikes. In setups with bidirectional charging capabilities, BEVs can also discharge stored energy during high-demand periods, further reducing a facility's net load and enhancing both economic performance and grid stability (Ostermann et al., 2022; Fenner et al., 2020; Liu et al., 2024).

Table 1

Key market characteristics relevant to the secondary use cases examined in this work. PS — peak shaving. SCO — self-consumption optimization. TOC — tariff-optimized charging. FCR — frequency containment reserve. aFRR — automatic frequency restoration reserve. SM-DA — spot market day-ahead. SM-IDA — spot market intraday auction. SM-IDC — spot market intraday continuous.

	PS	SCO	TOC	FCR ^d	aFRR ^g	SM-DA ^k	SM-IDA ^{n,o}	SM-IDC ^l
Location	On-site	On-site	On-site	Off-site	Off-site	Off-site	Off-site	Off-site
Remuneration	Power	Energy	Energy	Power	Power & energy	Energy	Energy	Energy
Time sectioning	–	–	Based on tariff	4h	4h ^h 15min ⁱ	15min ^l 1h ^m	15min	15min 30min 1h
Minimal bid/increment	–	–	–	1MW	1MW	0.1MW	0.1MW	0.1MW
Pricing	Tariff ^a	Tariff ^b	Tariff ^c	Pay-as-cleared ^e	Pay-as-cleared ^{e,h} , Pay-as-bid ^j	Pay-as-cleared ^e	Pay-as-cleared ^e	Pay-as-bid ^j
Gate closure time	–	–	–	08:00 (D ₁ ^f)	09:00 ^h (D ₁ ^f), 25min before delivery ⁱ	12:00 (D ₁ ^f)	15:00 (D ₁ ^f), 22:00 (D ₁ ^f), 10:00 (D ₀ ^p)	0min before delivery
Prequalification process	No	No	No	Yes	Yes	No	No	No

^aTariff is based on the network charges set by the local grid operator. ^bContract between direct marketer and PV system operator regarding feed-in tariff. ^cContract between energy supplier and fleet operator – common pricing strategies: time-of-use (TOU), critical peak pricing (CPP), real-time pricing (RTP). ^dCharacteristics given as examples for markets belonging to the European frequency containment reserve (FCR) Cooperation. ^eAll market participants receive the same clearing price, which is set by the highest accepted bid or the lowest accepted offer in the market. ^fDay before delivery. ^gCharacteristics given as examples for markets connected to the Platform for the International Coordination of the Automatic frequency restoration process and Stable System Operation. (PICASSO). ^hApplies for automatic frequency restoration reserve (aFRR) capacity market. ⁱApplies for aFRR energy market. ^jEach market participant is paid the exact price they bid. ^kCharacteristics given as examples for markets in the Multi-Regional-Coupling (MRC) framework. ^lSince 07/2025. ^mUntil 06/2025. ⁿCharacteristics given as examples for markets in the SIDC pan-European framework. ^oSince 06/2024. ^pDay of delivery. ^qCharacteristics given as examples for markets in Europe that run on the M7 trading system.

2.1.2. Self-consumption optimization

Self-consumption optimization (SCO) is an energy management strategy aimed at maximizing the on-site use of locally generated electricity, typically from renewable sources like solar systems (considered in this work). Reducing reliance on grid-supplied power can enhance energy efficiency and sustainability (Gomez-Cornejo et al., 2024). A key tactic involves shifting the charging and discharging schedules of BEVs: charging is typically scheduled during periods of peak solar generation, such as midday, while discharging aligns with high-demand periods of the site. The economic benefit stems from the gap between relatively high electricity purchase prices and the comparatively low feed-in tariffs for surplus PV energy, making it more cost-effective to consume generated power locally rather than exporting it to the public grid (Ganz et al., 2024; Rücker et al., 2020).

2.1.3. Tariff-optimized charging

Tariff-optimized charging (TOC) involves scheduling BEVs to charge during periods with lower electricity rates, based on time-varying tariff structures. Typical pricing schemes include Time-of-use (TOU) pricing, where rates differ across the day depending on peak and off-peak demand periods (Atthaporn et al., 2023; Huang et al., 2021; Biedenach and Ziemsky, 2022); Critical peak pricing (CPP), which imposes significantly higher rates during specific peak events to incentivize load reduction (Biedenach and Ziemsky, 2022; Yusuf et al., 2021); and Real-time pricing (RTP), where electricity prices are set on short notice, typically one day or one hour ahead, based on wholesale market fluctuations (Biedenach and Ziemsky, 2022; Amin et al., 2020). Other dynamic pricing models also exist, offering varying levels of responsiveness to grid conditions. By aligning charging behavior with lower-cost periods, this on-site use case can reduce electricity costs and contribute to grid stability by shifting demand away from peak times (Atthaporn et al., 2023).

2.2. Off-site

Off-site energy use cases refer to applications that operate in front of the utility meter, beyond the customer's premises. These use cases interact with the public energy system, enabling assets to participate in external markets or provide services to the grid. Unlike on-site strategies, which focus on internal cost savings, off-site use cases can generate revenue. By offering flexibility to the energy system, they contribute to grid stability, relieve network congestion, and respond to price signals in wholesale electricity markets. Moreover, they open up new business opportunities for BEV fleet operators that extend beyond direct energy cost savings at the depot level.

2.2.1. Balancing services provision

Participation in balancing services allows flexible energy assets to help maintain the real-time balance between electricity supply and demand in the grid. Transmission System Operators (TSOs) procure these services to ensure grid stability, especially as the share of variable renewable electricity sources (VRE) increases (Biancardi and Staffell, 2025; Sinsel et al., 2020). By rapidly adjusting power consumption or injection in response to grid needs, flexible assets can provide different types of balancing reserves, including frequency containment reserve (FCR), automatic frequency restoration reserve (aFRR), or manual frequency restoration reserve (mFRR). Participation in these markets requires meeting technical requirements such as response speed, accuracy, and availability, but offers revenue opportunities through capacity and activation payments (Blat Belmonte et al., 2023). Balancing services enable depot operators or fleet owners to monetize unused flexibility, such as controlled charging or discharging of BEVs, without disrupting core operations (Will and Ocker, 2024). When intelligently integrated into energy management systems, these services can contribute to grid stability while generating additional income streams beyond the primary energy needs of the site.

2.2.2. Trading in electricity spot markets

Spot market trading represents off-site use cases in which flexible energy assets participate in wholesale electricity markets operating close to real-time delivery. In contrast to balancing services, designed to ensure grid stability through ancillary service provision, spot market trading focuses on the economic optimization of electricity purchase and sale decisions based on short-term price signals. Market participants can engage in different trading windows, each offering varying lead times, from several hours ahead to just before delivery (Zachmann et al., 2023). The spot market day-ahead (SM-DA) enables participants to submit bids and offers for electricity delivery on the following day. Trading decisions in this window are typically based on price forecasts and expected load or generation profiles. Between the day-ahead and continuous intraday markets, the spot market intraday auction (SM-IDA) provides additional trading opportunities through discrete auctions held throughout the day. These auctions allow participants to adjust their positions based on updated market information, such as forecast revisions, while benefiting from a centralized price formation mechanism. In comparison, the spot market intraday continuous (SM-IDC) facilitates continuous trading closer to the time of delivery, allowing participants to fine-tune their positions in response to real-time updates, such as revised forecasts or unplanned deviations in

supply and demand. These markets are characterized by high price volatility and temporal granularity, creating opportunities for arbitrage (Zheng et al., 2025). By intelligently controlling the operation of flexible assets in response to these price signals, participants can capitalize on market dynamics. Thereby, BEV fleet operators may reduce energy costs or generate revenue by charging during low-price periods and discharging or shifting consumption away from high-price intervals, without compromising vehicle availability (Garg et al., 2023). The European Power Exchange Spot Market (EPEX SPOT) provides the platform for short-term electricity trading across multiple European countries. Furthermore, Papadimitriou et al. (2024) provide representative electricity price profiles for the European spot market day-ahead (SM-DA) and spot market intraday continuous (SM-IDC), based on historical time series from the EPEX SPOT.

2.2.3. Other emerging opportunities

In addition to the main off-site use cases assessed in this meta-analysis, several other applications could offer further potential to monetize BEV flexibility. These opportunities are not included in the present analysis because, to date, there are no or only very few quantitative assessments of their economic potential. While not yet supported by standardized market structures, these off-site opportunities may gain relevance as system needs grow and regulatory frameworks mature.

One notable application is congestion management, where BEVs help to alleviate grid constraints by modulating charging and discharging behavior in response to local signals from grid operators (Golab et al., 2025; Lehnhoff and Dorfner, 2025; RTE (Réseau de Transport d'Electricité), 2025; NODES, 2025; Ehrhart et al., 2025). Similarly, curtailment avoidance leverages BEVs to absorb excess renewable generation during periods of local oversupply, thereby reducing the need to curtail production (Saadatmandi et al., 2024; Park et al., 2022). Beyond these energy-balancing roles, BEVs with V2G capabilities may also support the black start process, contributing to system restoration following large-scale outages (Kaiss et al., 2025; Aguilar-Dominguez et al., 2022; Acharige et al., 2025). Moreover, as conventional synchronous generators are gradually phased out, BEVs could provide synthetic inertia and grid-forming services, helping to maintain frequency and voltage stability (Magdy et al., 2021; Wen et al., 2025). Finally, with the integration of advanced charging technologies, BEVs may be used for reactive power compensation, further supporting voltage regulation across the grid (Kaiss et al., 2025; Acharige et al., 2025; Chen et al., 2023).

Although these use cases are not part of the present meta-analysis, they illustrate the expanding role of BEVs in energy system integration and the potential for future value streams.

3. Methodology

The evaluation methodology follows a structured, multi-step approach to quantify the economic value of smart grid integration strategies across various secondary use cases. As shown in Table 1, the considered use cases and underlying markets differ in their remuneration mechanisms: some are compensated based on energy flexibility ($/(MWh)$), while others reward power flexibility ($/(MW)$). Based on this classification, the analyzed studies are grouped according to whether the reported economic potential value is linked to energy or power. As illustrated in Fig. 2, this categorization is the foundation for the methodological approach, enabling consistent normalization and comparison across studies with varying system assumptions, scales, and reporting formats.

Section 3.1 provides the selection criteria and meta-information on the considered studies. Furthermore, Section 3.2 explains the normalization approach to ensure comparability across these studies. The aFRR use case represents a special application, as it involves power- and energy-based remuneration, which is addressed in Section 3.3.

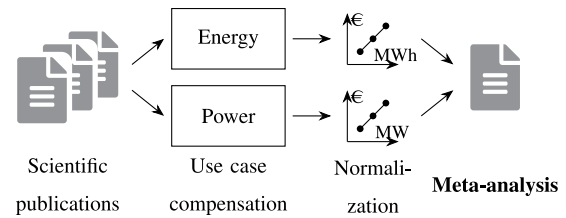


Fig. 2. Methodological overview of the study. Analyzed economic potential results from scientific publications on electric vehicle fleets operated in secondary use cases, categorized as either energy-compensated ($/(MWh)$) or power-compensated ($/(MW)$), are normalized to a common denominator for transfer and comparison.

3.1. Database

The literature database was compiled through a structured search and screening process. Relevant studies were identified using major scientific databases, including Scopus, Web of Science, and Google Scholar, based on combinations of keywords such as “battery electric vehicles”, “smart charging”, “vehicle-to-grid”, “flexibility”, “energy markets”, and “economic analysis”.

An initial set of publications was screened based on title and abstract to assess relevance to the economic evaluation of BEV flexibility. Subsequently, a full-text review was conducted to ensure that all included studies met the predefined inclusion criteria described below. Only peer-reviewed publications published from 2020 onwards were considered to reflect recent developments in market design, technology, and pricing structures.

Following this process, a total of 25 peer-reviewed publications were selected, meeting the following predefined inclusion criteria:

- Quantification of the economic potential of a single BEV or a BEV fleet in at least one of the considered use cases within European markets. This regional restriction ensures comparability across studies by limiting variations in market design, regulatory frameworks, and pricing structures.²
- Provision of information on flexibility capacity (both energy and power), vehicle type, aggregated fleet size, market location, quantification methodology, considered baseline scenario, and analysis period.
- Publication year of 2020 or later, to ensure the inclusion of recent and relevant findings.
- Optionally, provision of information on the reference year of the underlying electricity price data, including historical market observations or projected future market scenarios, as well as average vehicle idle times.

The reference year of the electricity price data varies across studies and reflects the assumptions adopted in the original publications. Future-oriented analyses typically rely on scenario-based electricity price projections derived from energy system simulations and electricity market models with high shares of renewable energy integration. These assumptions were retained as reported in the respective studies to preserve consistency with the underlying modeling frameworks. Consequently, uncertainties related to future electricity price developments and market evolution remain an inherent limitation of the underlying studies and, consequently, also of this meta-analysis.

² Including studies from fundamentally different regional market environments was deliberately avoided, as this would introduce additional heterogeneity in market design, pricing mechanisms, and regulatory conditions, thereby reducing the internal comparability of the meta-analysis.

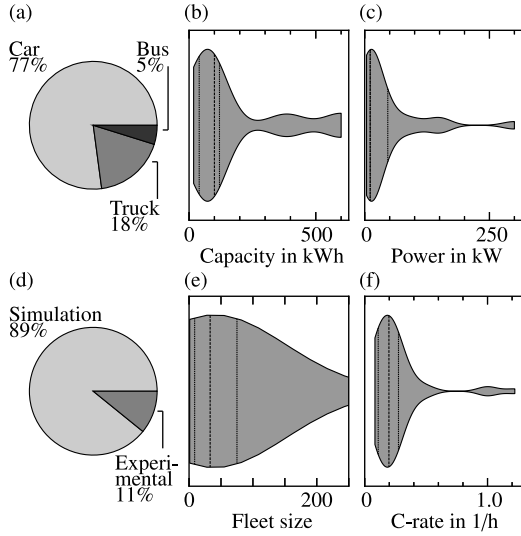


Fig. 3. Meta-information of analyzed 165 data points from scientific publications. (a) Share of fleet types; (b) Installed battery capacity per BEV; (c) Installed charger (dis-) charging powers per BEV; (d) Share of simulation-based and real-world field trials; (e) Fleet sizes — number of analyzed BEVs; (f) Employed maximum C-rates.

From the selected publications, a total of 165 individual scenario configurations across use cases, hereafter referred to as *data points*, are extracted. In this context, a data point is defined as a single scenario reported in the literature, representing the quantified economic outcome of a BEV or BEV fleet participating in a specific use case under a defined set of assumptions (e.g., fleet size, charging strategy, market conditions, and analysis period), and evaluated relative to an uncoordinated charging baseline. The included studies are summarized in Table A.1, categorized by secondary use case, publication year, analyzed markets, fleet type, evaluation methodology, and a brief description.

Fig. 3 provides an overview of the meta-information derived from the analyzed data points. A substantial share (77.1 %) relates to passenger cars, followed by trucks (18.1 %) and buses (4.8 %), indicating a strong emphasis on light-duty vehicle applications, likely due to their market maturity and broader data availability. Therefore, the associated battery capacities are mainly concentrated between 0 and 200 kWh. However, several data points linked to heavy-duty applications show capacities exceeding 400 kWh. Charging and discharging power values are generally below 150 kW, aligning with standard infrastructure for passenger vehicles and depot charging strategies in commercial fleet operations. Higher power levels are addressed only in a few cases, primarily in the context of heavy-duty applications.

While 89.2 % of the data points are based on simulation studies, only 10.8 % involve experimental or field trial data, underscoring the predominance of model-based analysis and highlighting a gap in empirical validation. Most data points consider relatively small fleets, often involving fewer than 50 aggregated BEVs. Large-scale scenarios with more than 200 BEVs are rare, likely due to limited deployment or restricted data access. The maximum C-rates, defined as the ratio of charging or discharging power to battery capacity, are predominantly below $0.5 \frac{1}{h}$, consistent with operational patterns typical of secondary use-case participation. Higher C-rates, usually associated with fast charging, where flexibility is rarely marketed, are absent. Overall, Fig. 3 shows that the analyzed data points are predominantly generated by simulation-based assessments of BEV fleets with varying battery capacities and charging power levels, with a notable emphasis on smaller vehicle platforms. The insights underline the motivation for the present study: to enable the transfer of economic potential estimates

across secondary use cases by establishing a consistent framework applicable to different fleet types and flexibility characteristics in terms of energy and power capacity.

3.2. Capacity-based normalization

Each reported annual value from the scientific literature on BEV fleets in secondary use cases is normalized to a per-unit basis. These values represent the annual nominal cost difference between the underlying smart charging strategy, where BEVs participate in flexibility markets while still fulfilling mobility requirements, and a baseline scenario characterized by uncoordinated charging.

In this context, the term *nominal* refers to modeled or estimated electricity cost values that exclude investment costs, inefficiencies (except charging/discharging efficiency), and other real-world financial factors. Furthermore, *nominal* indicates that the values are not adjusted for inflation and reflect the economic conditions at the time of analysis.

The normalization is performed relative to the underlying flexibility capacity: for energy-remunerated use cases, economic outcomes are expressed per unit of available battery energy capacity, whereas for power-remunerated use cases, values are expressed per unit of available power capacity (see Table 1). Accordingly, normalized results are expressed in units of €/MWh per year for energy-based use cases and €/MW per year for power-based use cases. This transformation preserves the economic value relative to the underlying flexibility capacity while removing the influence of fleet size and absolute system scale. As a result, it enables direct comparison of monetization potential across studies with differing technical configurations, market conditions, and aggregation levels.

Energy-based normalization. First, the flexible energy capacity E_{flex} available for market participation is defined based on the enabled battery capacity and the accessible SOC range for flexibility marketing:

$$E_{flex} = E_{bat} \cdot (SOC_{max}^{flex} - SOC_{min}^{flex}) \quad (1)$$

In Eq. (1), E_{bat} denotes the total battery energy capacity (Fig. 3(b)), while SOC_{max}^{flex} and SOC_{min}^{flex} represent the upper and lower limits of the SOC window that can be utilized under the respective smart charging control strategy.

The annual nominal value per flexible energy capacity is calculated in Eq. (2):

$$V_{nom,ann}^E = \frac{C_{nom,ann}^{flex} - C_{nom,ann}^{UC}}{E_{flex}} \quad (2)$$

Here, $C_{nom,ann}^{flex}$ refers to the nominal annual electricity costs under smart charging, and $C_{nom,ann}^{UC}$ denotes the nominal annual electricity costs under uncoordinated charging.

Power-based normalization. For use cases with remuneration based on available power capacity, the flexible power capacity P_{flex} is used for normalization. This value typically corresponds to the maximum charging or discharging power of the BEV fleet but may be limited in some studies for flexibility commercialization due to market participation requirements or operational constraints.

The annual nominal value per flexible power capacity is computed as:

$$V_{nom,ann}^P = \frac{C_{nom,ann}^{flex} - C_{nom,ann}^{UC}}{P_{flex}} \quad (3)$$

In Eq. (3), the numerator remains identical to that of Eq. (2), while the denominator P_{flex} denotes the fleet's usable charging or discharging power that is available for flexibility marketing.

Application. Both approaches - Eqs. (2) and (3) - enable a normalized representation of the economic benefit per unit of flexible capacity energy or power, respectively. The appropriate normalization is selected based on the remuneration logic of the respective secondary use case, highlighted in Table 1. This approach ensures consistent comparison across studies with heterogeneous modeling assumptions, system scales, and market designs.

Although the proposed normalization substantially improves cross-study comparability, several limitations remain. In particular, the normalization does not eliminate differences in underlying market structures, regulatory frameworks, optimization horizons, forecasting methodologies, or assumptions regarding battery degradation and charging infrastructure. Furthermore, the monetization potential reported in the analyzed studies is influenced by region-specific electricity price dynamics, flexibility availability patterns, and varying operational constraints. Consequently, the normalized values should not be interpreted as universally transferable economic indicators, but rather as standardized comparative metrics that enable a more consistent assessment of relative value potential across heterogeneous studies.

3.3. aFRR - split in energy and capacity remuneration

The aFRR use case represents a special case within the analyzed secondary applications, as it involves two distinct remuneration streams, according to Table 1: one for the provision of reserve capacity (power-based) and one for the activation of energy (energy-based) (Merten et al., 2020; Paredes and Aguado, 2024). Consequently, the economic value derived from aFRR participation comprises both capacity and energy market revenues.

However, some of the considered scientific publications for the database of this work do not report these components separately, instead providing only the total economic potential of aFRR participation. For consistent normalization, according to Eqs. (2) and (3), it is essential to distinguish between energy-based and power-based remuneration in order to apply the appropriate per-unit normalization.

To enable this differentiation in cases where only the aggregated revenue is reported, a splitting factor γ_{aFRR}^P is introduced. This factor represents the share of total revenue attributable to the capacity market, while its complement $\gamma_{aFRR}^E = (1 - \gamma_{aFRR}^P)$ corresponds to the share attributable to the energy market.

These factors are derived empirically from historical market data of the German aFRR system covering the years 2021 to 2024 (Regelleistung.net, 2025; Netztransparenz.de, 2025). Specifically, the analysis includes awarded reserve capacity prices (based on average submitted bid prices) and awarded volumes, as well as activated energy volumes and corresponding energy prices. For the energy component, the *minimum bid prices* published by the TSOs are used. This choice is motivated by the structure of the energy auction, in which each participant is paid based on their bid rather than a uniform market-clearing price. While average and marginal prices are also published, they tend to overestimate actual revenues: marginal prices are often skewed by intentionally high bids (e.g., up to 15 000 €/MWh) submitted by participants aiming to avoid activation, while these outliers likewise inflate average prices. In contrast, the minimum bid price reflects the lowest guaranteed payment and is a conservative but realistic basis for estimating revenue from energy activations.

The total remuneration for the capacity market over the considered period, denoted as R_{cap}^{total} , is calculated as the product of awarded capacity and respective clearing prices. Analogously, the total market remuneration from energy activations, R_{energy}^{total} , is obtained by multiplying the activated energy volumes with the corresponding energy prices and summing the values across all time intervals. All intermediate results and annual values used in calculating these totals are summarized in Table B.1 in Appendix B.

From these aggregated values, the splitting factors are determined as:

$$\gamma_{aFRR}^P = \frac{R_{cap}^{total}}{R_{cap}^{total} + R_{energy}^{total}}, \quad \gamma_{aFRR}^E = 1 - \gamma_{aFRR}^P \quad (4)$$

Once calculated, these factors are applied uniformly to all studies that report only the total annual nominal value of aFRR revenues, denoted $V_{nom,ann}^{aFRR}$. The total value is then decomposed into a power-based component:

$$V_{nom,ann}^P = \gamma_{aFRR}^P \cdot V_{nom,ann}^{aFRR} \quad (5)$$

and an energy-based component:

$$V_{nom,ann}^E = \gamma_{aFRR}^E \cdot V_{nom,ann}^{aFRR} = (1 - \gamma_{aFRR}^P) \cdot V_{nom,ann}^{aFRR} \quad (6)$$

This decomposition ensures methodological consistency across all analyzed data points and enables the standardized inclusion of aFRR-related results in the comparative assessment of energy- and power-based secondary use cases.

4. Results and discussion

The proposed methodology is applied to the secondary use cases in Table 1. Sections 4.1 and 4.2 present the results for energy- and power-remunerated use cases, respectively. Figs. 4 and 5 report the normalized annual nominal values $V_{nom,ann}^E$ and $V_{nom,ann}^P$ for unidirectional (light gray) and bidirectional (dark gray) integration; corresponding distribution statistics are provided in Tables C.1 and D.1.

Figs. 4 and 5 are designed as integrated result-interpretation figures. Beyond the distribution of normalized economic values (panel a), they incorporate key contextual variables (panels b–e), including price year, idle time, region, and vehicle type. This combined representation enables a direct linkage between economic outcomes and underlying scenario characteristics, supporting a more systematic interpretation of variability across studies.

Across all use cases, three key drivers consistently determine the achievable value: (i) market price spreads and tariff structures, (ii) the availability of idle time, particularly its temporal alignment with high-value periods, and (iii) bidirectionality, which expands the feasible set of flexibility options and enhances value capture. These drivers are reflected in the contextual distributions shown in panels (b)–(e), providing additional insight into the observed dispersion of results.

4.1. Economic potential of BEVs in energy-remunerated use cases

Energy-remunerated use cases include self-consumption optimization (SCO), tariff-optimized charging (TOC), trading on the spot market day-ahead (SM-DA), spot market intraday auction (SM-IDA), spot market intraday continuous (SM-IDC), and spot market consecutive (SM-con), as well as participation in the energy component of automatic frequency restoration reserve (aFRR).

On-site integration. On-site strategies show the lowest absolute values but benefit from low implementation complexity and no prequalification or direct market access requirements.

For self-consumption optimization (SCO), the results indicate a unidirectional median of 8.75 k€/MWh (IQR: 6.93–9.74), increasing to 11.68 k€/MWh with bidirectional operation (IQR: 10.42–13.80). This indicates stable savings potential with an additional upside from controlled discharge (e.g., building or site supply). Importantly, longer idle time alone does not guarantee higher value. This effect becomes apparent when linking the economic outcomes in Fig. 4(a) with the underlying scenario characteristics in Fig. 4(c), where higher values are associated with favorable temporal alignment between idle time and PV generation or demand peaks.

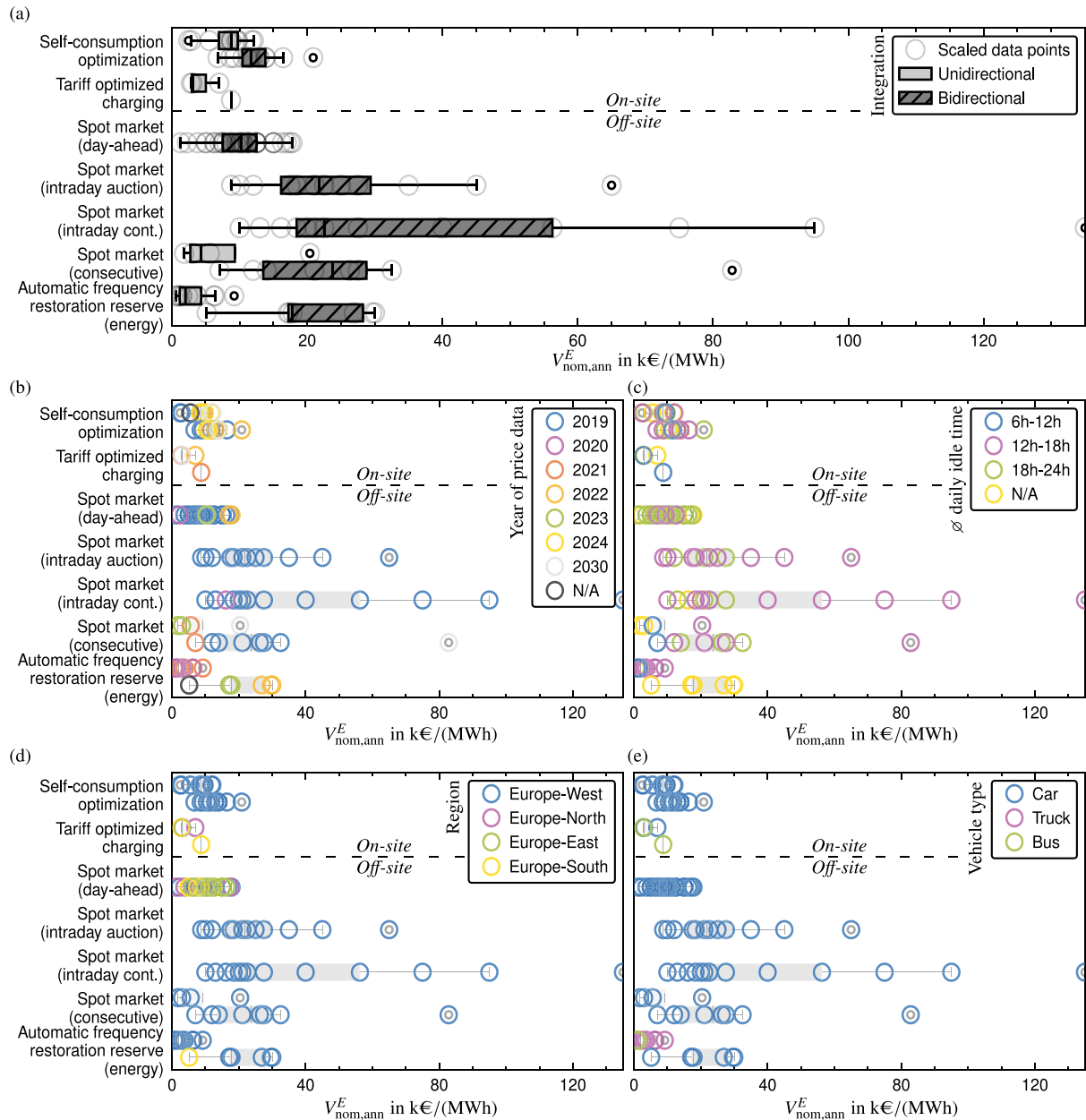


Fig. 4. Economic potential of aggregated electric vehicle batteries in secondary on-site and off-site use cases compensated for energy. (a) Distribution of normalized annual nominal values $V^E_{nom,ann}$ for unidirectional and bidirectional integration (median, IQR, Tukey whiskers). (b–e) Corresponding contextual information on underlying scenario characteristics: price year, average idle time, region, and vehicle type. Values are expressed in k€/MWh and represent cost advantages relative to uncoordinated charging.

tariff-optimized charging (TOC) yields lower values (unidirectional median 3.03 k€/MWh; IQR: 2.91–5.01). A single bidirectional result (8.73 k€/MWh) indicates potential upside but does not allow robust generalization. Since TOC tariffs are often linked to wholesale prices, its economic outcome is primarily determined by tariff structure and the fleet's ability to avoid charging during high-price windows.

Consequently, SCO can be characterized as a low-risk and low-complexity entry option with predictable but inherently capped savings potential, as value creation is constrained by local demand and generation profiles. In contrast, TOC exposes fleets more directly to wholesale

price dynamics and therefore requires sufficient temporal charging flexibility to consistently yield cost reductions.

Off-site integration. Off-site energy-remunerated markets exhibit clearly higher potential, especially under bidirectional integration, as they enable systematic energy arbitrage and participation in balancing energy activation.

For SM-DA, bidirectional integration yields a median of 10.18 k€/MWh (IQR: 7.50–12.50), reflecting comparatively stable day-ahead arbitrage returns. As illustrated in Fig. 4(d), performance is

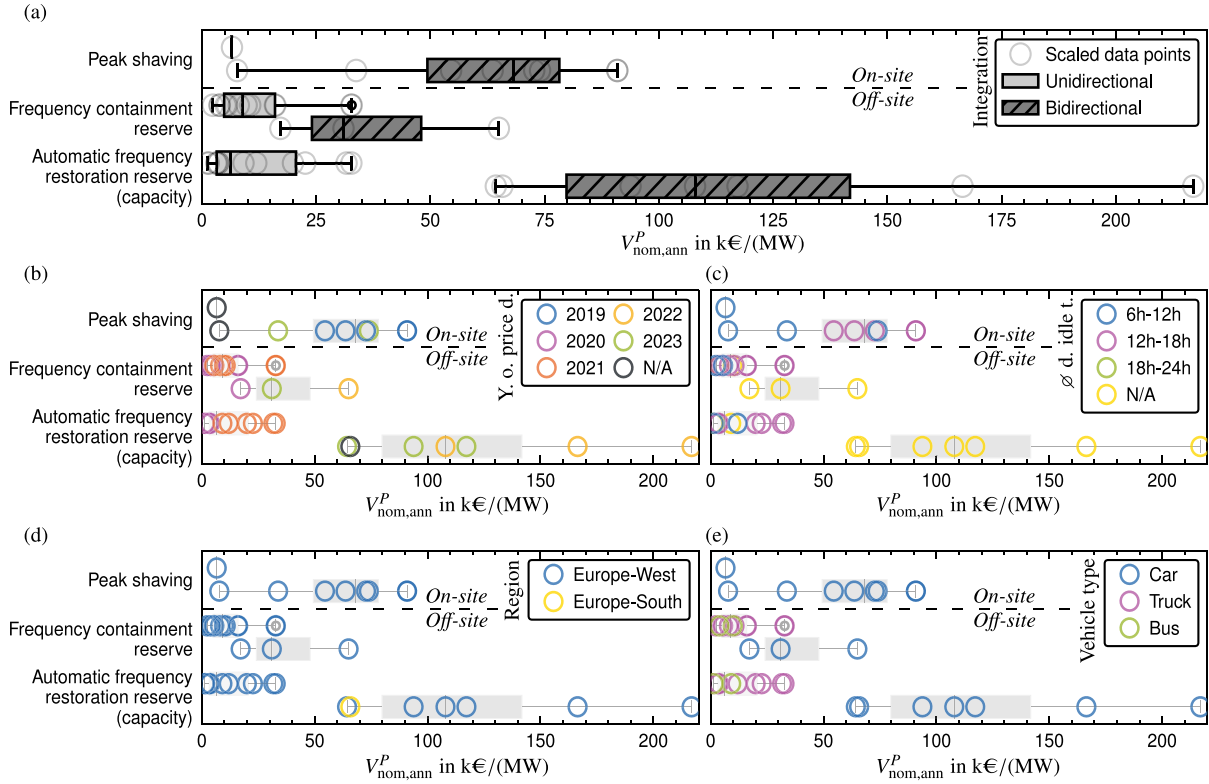


Fig. 5. Economic potential of aggregated electric vehicle batteries in secondary on-site and off-site use cases compensated for power. (a) Distribution of normalized annual nominal values $V_{nom,ann}^P$ for unidirectional and bidirectional integration (median, IQR, Tukey whiskers). (b–e) Corresponding contextual information on underlying scenario characteristics: price year, average idle time, region, and vehicle type. Values are expressed in k€/ (MW) and represent cost advantages relative to uncoordinated charging.

consistent across several countries, reflecting the harmonized rules established under the Single Day-ahead Coupling (SDAC) framework (ENTSO-E, 2025) and applied to many day-ahead markets in Europe. Furthermore, the price-year distribution in Fig. 4(b) indicates that the highest values are associated with 2022 market conditions. This perception is likely attributable to the correlation between European electricity market prices and natural gas price developments, which surged in 2022 due to geopolitical disruptions (Amer et al., 2025).

In contrast to stable day-ahead arbitrage returns, intraday trading provides substantially higher value: spot market intraday auction (SM-IDA) reaches a median of 21.78 k€/ (MWh) (IQR: 16.12–29.38), and SM-IDC reaches 22.50 k€/ (MWh) with a wider upper tail (IQR up to 56.25). The broader spread in continuous intraday trading is consistent with the higher temporal granularity and sensitivity to strategy design (forecasting, bidding logic) and to the alignment between fleet availability and high-price intervals.

SM-con (multi-market/consecutive participation) shows the highest median among trading-based use cases (23.68 k€/ (MWh); IQR: 13.50–28.75) and highlights the importance of bidirectionality: unidirectional SM-con remains modest (median 4.30 k€/ (MWh)). Overall, these results suggest that access to short-term price signals and the ability to both charge and discharge are key to capturing off-site arbitrage value.

Finally, aFRR(e) demonstrates that balancing activations can provide an additional and relatively stable energy-based revenue stream once access barriers are met. Bidirectional operation yields a median of 17.81 k€/ (MWh) (IQR: 17.21–28.25), whereas unidirectional participation remains low (median 1.98 k€/ (MWh)). As aFRR also includes

a capacity component (aFRR(p)), total aFRR revenues should be interpreted as the sum of energy- and power-based contributions. The relative importance of this power-based component is discussed in detail in Section 4.2.

Overall, the results reveal a clear hierarchy in energy-remunerated use cases. On-site applications provide stable but inherently limited economic potential, as value creation is constrained by local demand and generation patterns. In contrast, off-site markets, particularly intraday and multi-market participation, enable substantially higher revenues by leveraging short-term price volatility.

From a system perspective, this suggests that the economic value of BEV flexibility is primarily unlocked through exposure to external market signals rather than purely local optimization. Across all use cases, bidirectional operation emerges as a key enabler, significantly increasing value capture by expanding the range of feasible trading and balancing strategies. These findings highlight that both market access and operational flexibility are critical determinants of economic performance.

4.2. Economic potential of BEVs in power-remunerated use cases

Power-remunerated use cases include peak shaving (PS), frequency containment reserve (FCR), and automatic frequency restoration reserve (aFRR)(p).

On-site integration. PS is the only on-site power-remunerated case considered and monetizes avoided peak demand at the connection point. Due to limited unidirectional evidence (single data point), interpretation focuses on bidirectional cases, which reach a median of

68.18 k€/ (MW) (IQR: 49.35–78.23). Similar to on-site energy use cases, the key driver is not idle time per se but the alignment between vehicle availability and local demand peaks. Notably, the observed economic potential aligns with typical ranges of capacity-based network charges imposed by Distribution System Operators (DSOs), as exemplarily shown in Table D.2 for a German DSO. Depending on the voltage level and the annual number of full load hours (FLHs), capacity prices can range from below 20 k/(MW) to well above 150 k/(MW). In particular, sites with high FLHs face significantly higher capacity tariffs, which increases the value of avoiding peak demand through PS strategies. This tiered pricing structure may explain part of the spread observed in the bidirectional PS results, as use cases modeled under high-FLH conditions likely show greater economic benefit than those operating in low-FLH scenarios.

Off-site integration. For ancillary services, bidirectionality is the dominant value lever. In FCR, the unidirectional median is 8.95 k€/ (MW) (IQR: 4.88–16.01), while bidirectional operation increases the median to 31.00 k€/ (MW) (IQR: 24.09–48.00). Although bidirectional FCR is based on a small sample, the value increase is consistent across the available data.

The strongest power-based values occur in aFRR(p). Unidirectional participation remains limited (median 6.36 k€/ (MW); IQR: 3.25–20.61), while bidirectional integration reaches 107.92 k€/ (MW) (IQR: 79.79–141.82). This reflects the higher value of symmetric up- and downward reserve provision and indicates that aFRR(p) can become a major revenue pillar for fleets that meet qualification and operational requirements.

For both FCR and aFRR, the highest economic potentials are associated with the price year 2022, highlighting the strong sensitivity of balancing revenues to elevated system volatility and balancing demand Blat Belmonte et al. (2023).

In contrast to many energy-based applications, power-remunerated use cases exhibit high normalized values and strong sensitivity to bidirectional capability. Capacity-based markets such as FCR and aFRR reward the reliable provision of symmetric flexibility, making bidirectional operation an important prerequisite for capturing the highest value levels. Compared to energy arbitrage, these markets offer more stable revenue streams but require stricter qualification and operational constraints. Overall, the results suggest that power-based services can represent a major revenue pillar for BEV fleets, provided that technical and regulatory requirements are met.

Taken together, the results demonstrate that the economic potential of BEV fleet flexibility is highly dependent on both the type of use case and the degree of system integration. While on-site applications offer low-risk entry points with predictable savings, off-site participation, particularly in ancillary services, enables significantly higher normalized value creation.

From a market design perspective, this indicates that capacity-based mechanisms reward technically reliable and symmetric flexibility provision, thereby favoring bidirectional integration. However, this potential remains strongly conditioned by regulatory requirements, price volatility, and operational constraints. These findings underline the importance of aligning fleet operation strategies with specific market characteristics to fully exploit available flexibility.

4.3. Limitations and interpretation of significance

The findings enable a structured comparison across use cases, but several limitations affect absolute interpretation:

- **Normalization:** The per-unit normalization supports comparability but neglects cannibalization/saturation effects in limited-volume markets and does not represent optimized multi-use-case dispatch across markets. Therefore, values are most representative for small-scale participation.

- **Evidence base:** Most data points are simulation-based (89.2%), with limited field-trial evidence (10.8%), which may not fully capture operational uncertainty and implementation constraints.
- **Heterogeneity:** Studies differ in market years, regulatory settings, and regional conditions (e.g., network charge structures). Outliers may stem from specific price spikes and should not be generalized.
- **Small-*n* categories:** Some use cases (e.g., unidirectional PS, bidirectional TOC) are represented by few data points, limiting statistical robustness.

Hence, the results are best interpreted as robust *relative trends* and comparative opportunities across use cases and integration strategies rather than universally transferable absolute revenue levels.

5. Summary and conclusion

The intelligent integration of BEVs into the local and public energy system is a promising opportunity to reduce costs for fleet operators. While numerous studies have demonstrated the technical feasibility of BEV-based flexibility in both on-site and off-site applications, economic assessments have remained fragmented, lacking standardized metrics for cross-comparison. This meta-analysis addresses this gap by introducing the two normalization metrics $V_{\text{nom,ann}}^E$ and $V_{\text{nom,ann}}^P$, which provide a consistent and transferable framework to evaluate and compare the monetization potential of BEV fleet flexibility.

The meta-analysis covers 165 data points, each representing an individual economic potential analysis result, from 25 peer-reviewed publications. The analyzed secondary use cases include on-site strategies such as self-consumption optimization (SCO), tariff-optimized charging (TOC), and peak shaving (PS), as well as off-site participation in electricity spot markets (day-ahead and intraday) and ancillary service markets, notably frequency containment reserve (FCR) and automatic frequency restoration reserve (aFRR). The analysis revealed the following key insights:

- **On-site use cases** provide modest but stable economic value, with SCO yielding median revenues of up to 11.68 k/(MWh) for bidirectional integration. These use cases benefit from relatively low technical complexity but are limited in scalability due to their dependence on local site conditions and tariff structures.
- **Off-site use cases**, especially intraday spot market trading and participation in the aFRR market, demonstrate significantly higher value potential. Median revenues for bidirectional integration in these markets exceed 20 k/(MWh) (energy-based) and 100 k/(MW) (power-based), underscoring the economic relevance of enabling two-way power flows and participating in multiple market segments.
- **Bidirectional integration** consistently outperforms unidirectional strategies across all use cases, highlighting the strong influence of discharging capability on value creation. However, as this analysis focuses solely on operational cost, future assessments must also account for the capital expenditures associated with bidirectional charging infrastructure to provide a more complete economic evaluation.
- **Market price volatility** significantly shapes the economic potential of flexibility. Revenues tend to be higher in volatile markets (especially observable in analyses with market prices from 2021 to 2023) with price asymmetry and in scenarios where BEV fleets can respond rapidly to short-term signals or participate symmetrically in grid services.
- **Idle time and flexibility alignment** emerge as critical factors influencing economic performance, though long idle durations alone do not guarantee high revenues. Synchronization with the market or local generation and demand peaks is necessary to realize full potential.

Despite these findings, several limitations should be considered. The dominance of simulation-based data introduces uncertainty regarding real-world implementation. Additionally, the normalization approach does not capture saturation effects in limited markets or the economic potential of multi-use-case participation.

In addition to these limitations, the practical implementation of both on-site and off-site use cases depends on the availability of adequate information and communication technology (ICT) infrastructure. Realizing these applications requires reliable data acquisition and exchange, including information on vehicle SOC, availability, charging power, and user mobility constraints. Furthermore, coordinated control of distributed BEV fleets relies on robust communication interfaces between vehicles, charging infrastructure, local energy management systems, and external market platforms or aggregators.

Particularly for off-site applications, such as participation in electricity markets or ancillary services, low-latency communication, forecasting capabilities, and compliance with market-specific technical requirements are essential. These requirements may introduce additional investment costs, operational complexity, and cybersecurity considerations, which are typically not captured in the economic assessments reviewed in this study. While the analyzed studies implicitly assume the availability of such infrastructure, future research should more explicitly account for these enabling conditions to provide a more comprehensive evaluation of real-world feasibility.

In conclusion, this meta-analysis establishes a robust methodological foundation for the comparative economic assessment of BEV fleet flexibility in secondary energy use cases within Europe. Standardizing revenue metrics and applying them across diverse applications offers valuable insights for researchers, fleet operators, energy market actors, policymakers, and other industry experts. The results underscore that, intelligently integrated BEV fleets are not just transportation assets but valuable sources of system flexibility. By contributing to energy cost savings, smart and bidirectional system integration can reduce the total cost of ownership, making them increasingly attractive to fleet operators.

In future work, we will focus on bridging the gap between simulation-based projections and real-world implementation through empirical validation in a demonstration project with battery electric and bidirectional heavy-duty trucks. Comprehensive system-level analyses are needed to account for infrastructure investments, and the broader impact on fleet operations. Additionally, investigating the combined and coordinated use of multiple flexibility services, rather than assuming simple additive effects, can provide a more realistic understanding of market limitations and operational interactions. In principle, stacking on-site and off-site applications may further enhance the economic value of BEV flexibility. For example, BEVs could simultaneously support local peak shaving or self-consumption optimization while participating in external electricity markets or ancillary services during periods of available flexibility. However, realizing such multi-use-case strategies requires advanced energy management systems, prioritization mechanisms between competing services, and careful consideration of technical, operational, and regulatory constraints. These interactions are only partially addressed in the current literature and therefore represent an important direction for future research. Furthermore, the increasing demand for clean flexibility across Europe, quantified through the national assessments coordinated by EU Agency for the Cooperation of Energy Regulators (ACER) (ACER, 2025) and published in 2027, is expected to reveal new value streams. These developments could further enhance the attractiveness of marketing BEV fleet flexibility in future energy systems.

CRediT authorship contribution statement

Marcel Brödel: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Fabian Bussieweke:** Writing – review & editing.

Brian Dietermann: Writing – review & editing. **Niclas Klein:** Writing – review & editing. **Anna Paper:** Writing – review & editing. **Woan-Ho Park:** Writing – review & editing. **Philipp Rosner:** Writing – review & editing, Methodology. **Tom Winkler:** Writing – review & editing. **Markus Lienkamp:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Grammarly in order to improve readability and language of the work. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Considered scientific publications

See [Table A.1](#).

Appendix B. aFRR splitting factor

See [Table B.1](#).

Appendix C. Results - energy-remunerated use cases

See [Table C.1](#).

Appendix D. Results - power-remunerated use cases

See [Tables D.1](#) and [D.2](#).

Data availability

The source code and data are available on GitHub: https://github.com/TUMFTM/review_monetizing_bev_flexibility.

Table A.1

Summary of analyzed scientific publications for this work. The table provides an overview of the reviewed publications categorized by use case, publication year, analyzed markets, fleet type, evaluation methodology, and a brief description.

Use case	Publication (year)	Market (ISO 3166)	Fleet type	Meth.	Description
PS	Borlaug et al. (2021)	US	Truck	Sim.	The publication analyzes three heavy-duty truck fleets charging at the depot, focusing on the impact of peak shaving strategies on the electricity distribution system. Therefore, depot charging load profiles are applied to 36 distribution real-world substations.
PS	Meiers and Frey (2024)	DE	Car	Sim.	The publication assesses uni- and bidirectional charging strategies for two BEV fleets and two comparable configurations of stationary battery storage systems to explore their economic potential for peak shaving on the Saarland University campus.
PS	Al-Hanahi et al. (2022)	CA	Truck	Sim.	The publication explores smart charging using a proposed system designed for unidirectional truck fleets operating at a commercial facility to minimize the peak demand of the facility's aggregate load profile.
PS	van Krieking et al. (2021)	BE	Car	Sim.	The publication presents two BEV charging schedulers based on model predictive control algorithms, applied to four distinct charging strategies aimed at peak shaving. A simulation is conducted for a charging site in Brussels.
PS	Weiß et al. (2021)	DE	Car	Sim.	The publication examines the effects of unidirectional and bidirectional BEV fleet load management strategies on the distribution grid load through multiple case studies.
PS	Prakash et al. (2022)	AU	Truck	Sim.	The publication proposes a framework for optimal charging station placement and BEV scheduling to reduce peak load and congestion in low-voltage networks, accounting for uncertainties in BEV usage and grid behavior.
PS	Kern and Bukhari (2021)	DE	Car	Sim.	The publication evaluates the cost-saving potential of smart and bidirectional BEV charging for peak shaving in Germany across various fleet sizes.
SCO	Ganz et al. (2024)	DE	Car	Sim.	The publication assesses the impact of self-consumption optimization using uni- and bidirectional BEVs on the 2030 energy system. An energy system model simulates diverse population groups with varying mobility behaviors to achieve this.
SCO	Vollmuth et al. (2024)	DE	Car	Sim.	The publication assesses the financial benefits for private BEV owners in Germany using uni- and bidirectional charging to optimize PV self-consumption, based on the <i>eflame</i> model.
SCO	Kern et al. (2022)	DE	Car	Sim.	The publication evaluates the revenue potential for a German household using unmanaged, smart, and bidirectional BEV charging to optimize PV self-consumption.
SCO	von Bonin et al. (2022)	DE	Car	Sim.	The publication examines how PV optimized charging affects grid load and household economics, based on a sample of 100 BEV-owning households in a metropolitan area.
SCO	Rücker et al. (2020)	DE	Car	Sim.	The publication investigates smart charging strategies for a BEV at home and workplace with a PV system. In addition to evaluating economic charging objectives, it examines the effects on self-consumption, self-sufficiency rates, and battery aging.
TOC	von Bonin et al. (2022)	DE	Car	Sim.	The publication examines how dynamic electricity prices affect charging behavior, grid load, and household economics, based on a sample of 100 BEV-owning households in a metropolitan area.
TOC	Manzolini et al. (2022)	PT	Bus	Sim.	The publication presents a case study based on real-world data from a BEV fleet in a medium-sized Portuguese city to assess tariff-optimized uni- and bidirectional charging.
TOC	Einolander et al. (2023)	FI	Car	Sim.	The publication evaluates the economic benefits of V2H technology for detached households with different primary heating sources under a dynamic electricity tariff, using real-world data within the Helsinki region.

(continued on next page)

Table A.1 (continued).

Use case	Publication (year)	Market (ISO 3166)	Fleet type	Meth.	Description
FCR	Will and Ocker (2024)	DE	Bus & Truck	Sim.	The publication analyzes trucks' unidirectional FCR flexibility potential across five typical use cases and city buses in Germany. The resulting additional revenues could help lower electricity costs for depot operators.
FCR	Tepe et al. (2022)	AT, BE, CH, CZ, DE, DK, FR, NL, SI,	Car	Sim.	The publication analyzes the potential of commercial BEVs to provide balancing power on Central Europe's FCR market. Optimization methods are used to form optimal pooling combinations based on the vehicles' power and energy capability profiles.
FCR	Piffaretti et al. (2024)	CH	Car	Exp.	The publication presents the final results of a Swiss research project involving 50 bidirectional BEVs used for car sharing, deployed across 39 charging locations nationwide, demonstrating their flexibility through participation in the FCR market.
aFRR	Will and Ocker (2024)	DE	Bus & Truck	Sim.	The publication analyzes trucks' unidirectional aFRR flexibility potential across five typical use cases and city buses in Germany. The resulting additional revenues could help lower electricity costs for depot operators.
aFRR	Piffaretti et al. (2024)	CH	Car	Exp.	The publication presents the final results of a Swiss research project involving 50 bidirectional BEVs used for car sharing, deployed across 39 charging locations nationwide, demonstrating their flexibility through participation in the aFRR market.
aFRR	de la Torre et al. (2023)	ES	Car	Sim.	The publication introduces an optimization model for BEV aggregators, accounting for key profitability factors. A case study illustrates potential earnings for BEV owners providing aFRR and other services.
SM-DA	Vollmuth et al. (2024)	DE	Car	Sim.	The publication assesses the financial benefits for private BEV owners in Germany using uni- and bidirectional charging to participate in SM-DA energy arbitrage trading, based on the <i>eFlame</i> model.
SM-DA	Tepe et al. (2022)	DE	Car	Sim.	The publication analyzes the potential of commercial BEVs to participate in SM-DA trading on the EPEX SPOT. Optimization methods are used to form optimal pooling combinations based on the vehicles' power and energy capability profiles.
SM-DA	Piffaretti et al. (2024)	CH	Car	Exp.	The publication presents the final results of a Swiss research project involving 50 bidirectional BEVs used for car sharing, deployed across 39 charging locations nationwide, demonstrating their flexibility through participation in SM-DA trading.
SM-DA	Kern et al. (2020)	AT, BE, BG, CH, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IT, LT, LV, NL, NO, PL, PT, RO, RS, SE, SI, SK	Car	Sim.	The publication investigates the revenue potential of various bidirectional BEV pools, reflecting different user behaviors, in the SM-DA market across several European countries.
SM-DA	Lesjak (2023)	DE	Car	Sim.	This publication examines two residential BEV fleets in Germany, consisting of 50 and 500 vehicles. It presents an optimization model to minimize costs, assuming perfect foresight on energy prices and consumption.
SM-IDA	Vollmuth et al. (2024)	DE	Car	Sim.	The publication assesses the financial benefits for private BEV owners in Germany using uni- and bidirectional charging to participate in SM-IDA energy arbitrage trading, based on the <i>eFlame</i> model.
SM-IDA	Kern et al. (2020)	DE	Car	Sim.	The publication investigates the revenue potential of various bidirectional BEV pools, reflecting different user behaviors, in the SM-IDA market.
SM-IDA	Teske et al. (2021)	DE	Car	Sim.	The publication investigates the revenue potential of three bidirectional BEV scenarios, each characterized by different user parking behaviors, in the SM-IDA market.
SM-IDC	Vollmuth et al. (2024)	DE	Car	Sim.	The publication assesses the financial benefits for private BEV owners in Germany using uni- and bidirectional charging to participate in SM-IDC energy arbitrage trading, based on the <i>eFlame</i> model.
SM-IDC	Tepe et al. (2022)	DE	Car	Sim.	The publication analyzes the potential of commercial BEVs to participate in SM-IDC trading on the EPEX SPOT. Optimization methods are used to form optimal pooling combinations based on the vehicles' power and energy capability profiles.
SM-IDC	Kern et al. (2020)	DE	Car	Sim.	The publication investigates the revenue potential of various bidirectional BEV pools, reflecting different user behaviors, in the SM-IDC market.
SM-IDC	Teske et al. (2021)	DE	Car	Sim.	The publication investigates the revenue potential of three bidirectional BEV scenarios, each characterized by different user parking behaviors, in the SM-IDC market.
SM-IDC	Chemudupaty et al. (2025)	DE	Car	Sim.	The publication presents trading strategies for managing the power demand of a 1000-BEV fleet while leveraging additional flexibility in the SM-IDC market. Usage uncertainty is addressed through a dedicated flexibility forecasting algorithm.

Table B.1

Calculated splitting factors γ_{aFRR}^P , γ_{aFRR}^E based on the German aFRR market. The table presents yearly values for mean upward (↑) and downward (↓) capacity and energy prices, total market remunerations, and the corresponding splitting factors from 2021 to 2024. Price data is available in [Regelleistung.net](https://www.regelleistung.net) (2025). Activated balancing capacity data is available in [Netztransparenz.de](https://www.netztransparenz.de) (2025).

Metric	Unit	2021	2022	2023	2024
∅ cap. price ↑ (avg.)	€/ (MW)	11.60	13.39	14.61	13.14
∅ cap. price ↓ (avg.)	€/ (MW)	10.43	13.66	17.92	10.84
∅ en. price ↑ (min.)	€/ (MWh)	135.35	427.75	143.46	102.49
∅ en. price ↓ (min.)	€/ (MWh)	45.43	87.09	81.39	66.16
∑ cap. remuneration ↑	M€	212.92	233.69	248.65	226.83
∑ cap. remuneration ↓	M€	181.17	228.67	291.06	176.57
$R_{\text{cap}}^{\text{total}}$	M€	394.09	462.36	539.71	403.39
∑ en. remuneration ↑	M€	131.60	112.63	90.88	62.17
∑ en. remuneration ↓	M€	33.63	33.35	91.06	41.92
$R_{\text{energy}}^{\text{total}}$	M€	165.22	145.98	181.94	104.09
γ_{aFRR}^P	–	0.7046	0.7600	0.7479	0.7949
γ_{aFRR}^E	–	0.2954	0.2400	0.2521	0.2051

Table C.1

Statistical summary of economic potential for energy-remunerated use cases. The table reports descriptive distribution metrics (Min, Q1, Median, Q3, Max) of the annual nominal value $V_{\text{nom,ann}}^E$ for each use case, shown separately for cases without (X) and with (✓) bidirectionality. All values are expressed in k/(MWh). Entries marked with ‘–’ indicate unavailable data.

Use case	Bidi.	Min	Q1 (25%)	Median	Q3 (75%)	Max
SCO	X	2.43	6.93	8.75	9.74	12.09
	✓	6.74	10.42	11.68	13.80	20.88
TOC	X	2.80	2.91	3.03	5.01	6.98
	✓	8.73	8.73	8.73	8.73	8.73
SM-DA	X	–	–	–	–	–
	✓	1.25	7.50	10.18	12.50	17.80
SM-IDA	X	–	–	–	–	–
	✓	8.75	16.12	21.78	29.38	65.00
SM-IDC	X	–	–	–	–	–
	✓	10.00	18.42	22.50	56.25	135.00
SM-con	X	1.79	2.71	4.30	9.29	20.38
	✓	7.01	13.50	23.68	28.75	82.82
aFRR(e)	X	0.63	1.14	1.98	4.31	9.19
	✓	5.13	17.21	17.81	28.25	30.02

Table D.1

Statistical summary of economic potential for power-remunerated use cases. The table reports descriptive distribution metrics (Min, Q1, Median, Q3, Max) of the annual nominal value $V_{\text{nom,ann}}^P$ for each use case, shown separately for cases without (X) and with (✓) bidirectionality. All values are expressed in k/(MW).

Use case	Bidi.	Min	Q1 (25%)	Median	Q3 (75%)	Max
PS	X	6.59	6.59	6.59	6.59	6.59
	✓	7.73	49.35	68.18	78.23	90.91
FCR	X	2.39	4.88	8.95	16.01	32.80
	✓	17.18	24.09	31.00	48.00	65.00
aFRR(p)	X	1.30	3.25	6.36	20.61	32.72
	✓	64.32	79.79	107.92	141.82	216.93

Table D.2

Network charges by grid voltage level and full load hours across years for a German DSO ([Bayernwerk](https://www.bayernwerk.de), 2025). The table presents annual values for capacity network charges, differentiated by voltage level and whether annual FLHs are below or above 2500. All values are expressed in €/kW (= k/(MW)).

Voltage Level	FLH	2021	2022	2023	2024	2025
High	<2,500 h	11.80	11.82	12.45	21.15	22.14
Medium	<2,500 h	13.51	13.69	16.72	23.62	23.99
Low	<2,500 h	15.04	15.48	18.11	25.15	21.93
High	≥2,500 h	93.90	97.26	109.41	164.85	175.16
Medium	≥2,500 h	123.38	125.31	142.28	190.80	182.79
Low	≥2,500 h	97.75	95.77	118.22	157.29	156.60

References

- ACER, 2025. ACER approves EU-wide methodology to assess national electricity flexibility needs. URL <https://www.ffe.de/en/publications/german-electricity-prices-on-epex-spot-2024/>.
- Acharige, S.S., Haque, M.E., Arif, M.T., Hosseinzadeh, N., Hasan, K.N., Hossain, M.J., Muttaqi, K.M., 2025. Grid integration of electric vehicles - impact assessment and remedial measures. *J. Power Sources* 650, 236697. <http://dx.doi.org/10.1016/j.jpowsour.2025.236697>.
- Aguilar-Dominguez, D., Ejeh, J., Brown, S.F., Dunbar, A.D., 2022. Exploring the possibility to provide black start services by using vehicle-to-grid. *Energy Rep.* 8, 74–82. <http://dx.doi.org/10.1016/j.egy.2022.06.111>.
- Al-Hanahi, B., Ahmad, I., Habibi, D., Masoum, M.A.S., 2021. Charging infrastructure for commercial electric vehicles: Challenges and future works. *IEEE Access* 9, 121476–121492. <http://dx.doi.org/10.1109/ACCESS.2021.3108817>.
- Al-Hanahi, B., Ahmad, I., Habibi, D., Masoum, M.A., 2022. Smart charging strategies for heavy electric vehicles. *ETransportation* 13, 100182. <http://dx.doi.org/10.1016/j.etrans.2022.100182>.
- Aldahmashi, J., Ma, X., 2024. Real-time energy management in smart homes through deep reinforcement learning. *IEEE Access* 12, 43155–43172. <http://dx.doi.org/10.1109/ACCESS.2024.3375771>.
- Aliasghari, P., Mohammadi-Ivatloo, B., Abapour, M., 2020. Risk-based scheduling strategy for electric vehicle aggregator using hybrid stochastic/IGDT approach. *J. Clean. Prod.* 248, 119270. <http://dx.doi.org/10.1016/j.jclepro.2019.119270>.
- Almughram, O., Abdullah ben Slama, S., Zafar, B.A., 2023. A reinforcement learning approach for integrating an intelligent home energy management system with a vehicle-to-home unit. *Appl. Sci.* 13 (9), 5539. <http://dx.doi.org/10.3390/app13095539>.
- Amer, A.A., Shaban, K., Massoud, A.M., 2023. DRL-HEMS: Deep reinforcement learning agent for demand response in home energy management systems considering customers and operators perspectives. *IEEE Trans. Smart Grid* 14 (1), 239–250. <http://dx.doi.org/10.1109/TSG.2022.3198401>.
- Amer, N., Wasmeier, L., von Roon, S., Kern, T., 2025. German electricity prices on EPEX spot 2024. URL <https://www.ffe.de/en/publications/german-electricity-prices-on-epex-spot-2024/>.
- Amin, A., Tareen, W.U.K., Usman, M., Ali, H., Bari, I., Horan, B., Mekhilef, S., Asif, M., Ahmed, S., Mahmood, A., 2020. A review of optimal charging strategy for electric vehicles under dynamic pricing schemes in the distribution charging network. *Sustainability* 12 (23), 10160. <http://dx.doi.org/10.3390/su122310160>.
- Andersen, P.B., Sousa, T., Thingvad, A., Berthou, L.S., Kulahci, M., 2018. Added value of individual flexibility profiles of electric vehicle users for ancillary services. In: 2018 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm). pp. 1–6. <http://dx.doi.org/10.1109/SmartGridComm.2018.8587585>.
- Atthaporn, S., Charoenlarnnopparut, C., Suksumpong, P., Kudtongngam, J., 2023. An optimized charging for EV using numerical method in consideration of TOU. In: 2023 IEEE PES 15th Asia-Pacific Power and Energy Engineering Conference. APPEEC, IEEE, pp. 1–5. <http://dx.doi.org/10.1109/APPEEC57400.2023.10562024>.
- Aznavi, S., Fajri, P., Asrari, A., Harirchi, F., 2020. Realistic and intelligent management of connected storage devices in future smart homes considering energy price tag. *IEEE Trans. Ind. Appl.* 56 (2), 1679–1689. <http://dx.doi.org/10.1109/TIA.2019.2956718>.
- Bayernwerk, 2025. Netzentgelte archiv. URL <https://www.bayernwerk-netz.de/de/bayernwerk-netz-gmbh/netzinformation/netzentgelte/netzentgelte-strom.html>.
- Bhoir, S., Caliendo, P., Brivio, C., 2021. Impact of V2G service provision on battery life. *J. Energy Storage* 44, 103178. <http://dx.doi.org/10.1016/j.est.2021.103178>.
- Biancardi, A., Staffell, I., 2025. How do electricity TSOs embrace innovation to future-proof their role in the energy transition? *Energy Strat. Rev.* 59, 101746. <http://dx.doi.org/10.1016/j.est.2025.101746>.
- Biednach, F., Ziemsky, V., 2022. Opportunity or risk? Model-based optimization of electric vehicle charging costs for different types of variable tariffs from a consumer perspective. *IET Conf. Proc.* 2022 (3), 883–887. <http://dx.doi.org/10.1049/icp.2022.0839>.
- Blat Belmonte, B., Mouratidis, P., Franke, G., Rinderknecht, S., 2023. Developments in the cost of grid balancing services and the design of the European balancing market. *Energy Rep.* 10, 910–931. <http://dx.doi.org/10.1016/j.egy.2023.07.045>.
- Blonsky, M., McKenna, K., Maguire, J., Vincent, T., 2022. Home energy management under realistic and uncertain conditions: A comparison of heuristic, deterministic, and stochastic control methods. *Appl. Energy* 325, 119770. <http://dx.doi.org/10.1016/j.apenergy.2022.119770>.
- von Bonin, M., Dörre, E., Al-Khrouz, H., Braun, M., Zhou, X., 2022. Impact of dynamic electricity tariff and home PV system incentives on electric vehicle charging behavior: Study on potential grid implications and economic effects for households. *Energies* 15 (3), 1079. <http://dx.doi.org/10.3390/en15031079>.
- Borlaug, B., Muratori, M., Gilleran, M., Woody, D., Muston, W., Canada, T., Ingram, A., Gresham, H., McQueen, C., 2021. Heavy-duty truck electrification and the impacts of depot charging on electricity distribution systems. *Nat. Energy* 6 (6), 673–682. <http://dx.doi.org/10.1038/s41560-021-00855-0>.
- Bovera, F., Rancilio, G., 2024. The role of implicit and explicit economic signals for flexibility provision by EV aggregates: technical evidence and policy recommendations. *Incent. Digit. Flex. Green Transit.* URL <https://hdl.handle.net/11311/1273022>.
- Brinkel, N., Visser, L., van Sark, W., Alskaf, T., 2023. A novel forecasting approach to schedule aggregated electric vehicle charging. *Energy AI* 14, 100297. <http://dx.doi.org/10.1016/j.egyai.2023.100297>.
- Chemudupaty, R., Hornek, T., Pavić, I., Potenciano Menci, S., 2025. Optimizing trading of electric vehicle charging flexibility in the continuous intraday market under user and market uncertainties. *Appl. Energy* 381, 125103. <http://dx.doi.org/10.1016/j.apenergy.2024.125103>.
- Chen, J., Bei, G., Zhang, Q., Huang, L., Liu, K., Chen, P., 2023. Integrated reactive power optimization for distribution systems considering electric vehicle dis-/charging support. *Energy Rep.* 9, 1888–1896. <http://dx.doi.org/10.1016/j.egy.2023.04.189>.
- Chen, Q., Liu, N., Lu, X., Zhang, J., 2014. A heuristic charging strategy for real-time operation of PV-based charging station for electric vehicles. In: 2014 IEEE Innovative Smart Grid Technologies - Asia (ISGT ASIA). IEEE, pp. 465–469. <http://dx.doi.org/10.1109/ISGT-Asia.2014.6873836>.
- Dioha, M.O., Ruggles, T.H., Ashfaq, S., Caldeira, K., 2022. Idealized analysis of relative values of bidirectional versus unidirectional electric vehicle charging in deeply decarbonized electricity systems. *IScience* 25 (9), 104906. <http://dx.doi.org/10.1016/j.isci.2022.104906>.
- Ehrhart, K.-M., Eicke, A., Hirth, L., Ocker, F., Ott, M., Schlecht, I., Wang, R., 2025. Analysis of a capacity-based redispatch mechanism. *Energy Econ.* 149, 108751. <http://dx.doi.org/10.1016/j.eneco.2025.108751>.
- Einolander, J., Kiviahio, A., Lahdelma, R., 2023. Household electricity cost optimization with vehicle-to-home technology and mixed-integer linear programming. In: 2023 International Conference on Future Energy Solutions. FES, IEEE, pp. 1–5. <http://dx.doi.org/10.1109/FES57669.2023.10182713>.
- ENTSO-E, 2025. Single day-ahead coupling (sdac). URL https://www.entsoe.eu/network_codes/cacm/implementation/sdac/#market-information.
- Fenner, P., Rauma, K., Rautiainen, A., Supponen, A., Rehtanz, C., Järventausta, P., 2020. Quantification of peak shaving capacity in electric vehicle charging – findings from case studies in Helsinki Region. *IET Smart Grid* 3 (6), 777–785. <http://dx.doi.org/10.1049/iet-stg.2020.0001>.
- Figgner, J., Tepe, B., Rücker, F., Schoeneberger, I., Hecht, C., Jossen, A., Sauer, D.U., 2022. The influence of frequency containment reserve flexibilization on the economics of electric vehicle fleet operation. *J. Energy Storage* 53, 105138. <http://dx.doi.org/10.1016/j.est.2022.105138>.
- Forootani, A., Rastegar, M., Jooshaki, M., 2022. An advanced satisfaction-based home energy management system using deep reinforcement learning. *IEEE Access* 10, 47896–47905. <http://dx.doi.org/10.1109/ACCESS.2022.3172327>.
- Ganz, K., Kern, T., Hinterstocker, M., 2024. Systemic evaluation of PV self-consumption optimization using electric vehicles. *World Electr. Veh. J.* 15 (3), 98. <http://dx.doi.org/10.3390/wevj15030098>.
- Garg, K., Misra, P., Bichpuriya, Y., Vasan, A., 2023. Managed bidirectional charging of electric fleets with arbitrage. In: The Third International Conference on Artificial Intelligence and Machine Learning Systems. ACM, New York, NY, USA, pp. 1–9. <http://dx.doi.org/10.1145/3639856.3639860>.
- Ghatikar, G., Alam, M.S., 2023. Technology and economics of electric vehicle power transfer: insights for the automotive industry. *Energy Informatics* 6 (1), <http://dx.doi.org/10.1186/s42162-023-00300-4>.
- Golab, A., Loschan, C., Zwickl-Bernhard, S., Auer, H., 2025. The value of flexibility of commercial electric vehicle fleets in the redispatch of congested transmission grids. *Energy* 316, 134385. <http://dx.doi.org/10.1016/j.energy.2025.134385>.
- Gomez-Cornejo, J., Lopez, I., Aranzabal, I., Alberdi, R., Erezuma, E., 2024. Algorithm for managing the consumption curve of a consumer with self-consumption photovoltaic generation. *Renew. Energies, Environ. Power Qual. J.* 2, 148–154. <http://dx.doi.org/10.24084/reepqj24.325>.
- Heilmann, C., Friedl, G., 2021. Factors influencing the economic success of grid-to-vehicle and vehicle-to-grid applications—A review and meta-analysis. *Renew. Sustain. Energy Rev.* 145, 111115. <http://dx.doi.org/10.1016/j.rser.2021.111115>.
- Hj Osman, M.S., Wai Shin, H., Diyanah Nizamuddin, A., Ab Muis, Z., Keng Yinn, W., Huiyi, T., 2024. Vehicle to building (v2b) peak load shaving and tariff analysis. *IOP Conf. Ser.: Earth Environ. Sci.* 1395 (1), 012019. <http://dx.doi.org/10.1088/1755-1315/1395/1/012019>.
- Huang, J., Wang, X., Wang, Y., Ma, Z., Chen, X., Zhang, H., 2021. Charging navigation strategy of electric vehicles considering time-of-use pricing. In: 2021 6th Asia Conference on Power and Electrical Engineering. ACPEE, IEEE, pp. 715–720. <http://dx.doi.org/10.1109/ACPEE51499.2021.9436864>.
- Huy, T.H.B., Truong Dinh, H., Ngoc Vo, D., Kim, D., 2023. Real-time energy scheduling for home energy management systems with an energy storage system and electric vehicle based on a supervised-learning-based strategy. *Energy Convers. Manage.* 292, 117340. <http://dx.doi.org/10.1016/j.enconman.2023.117340>.
- Jankowiak, C., Zacharopoulos, A., Brandoni, C., Keatley, P., MacArtain, P., Hewitt, N., 2020. Assessing the benefits of decentralised residential batteries for load peak shaving. *J. Energy Storage* 32, 101779. <http://dx.doi.org/10.1016/j.est.2020.101779>.

- Kaiss, M., Wan, Y., Gebbran, D., Vila, C.U., Dragičević, T., 2025. Review on virtual power plants/virtual aggregators: Concepts, applications, prospects and operation strategies. *Renew. Sustain. Energy Rev.* 211, 115242. <http://dx.doi.org/10.1016/j.rser.2024.115242>.
- Kern, T., Bukhari, B., 2021. Peak shaving – a cost-benefit analysis for different industries.
- Kern, T., Dossow, P., Morlock, E., 2022. Revenue opportunities by integrating combined vehicle-to-home and vehicle-to-grid applications in smart homes. *Appl. Energy* 307, 118187. <http://dx.doi.org/10.1016/j.apenergy.2021.118187>.
- Kern, T., Dossow, P., von Roon, S., 2020. Integrating bidirectionally chargeable electric vehicles into the electricity markets. *Energies* 13 (21), 5812. <http://dx.doi.org/10.3390/en13215812>.
- de la Torre, S., Aguado, J.A., Sauma, E., 2023. Optimal scheduling of ancillary services provided by an electric vehicle aggregator. *Energy* 265, 126147. <http://dx.doi.org/10.1016/j.energy.2022.126147>.
- Lehnhoff, S., Dorfner, J., 2025. Redispatch 3.0. URL <https://www.bmwk.de/Redaktion/EN/Artikel/Digital-World/GAIA-X-Use-Cases/redispatch-30.html>.
- Lei, X., Zhong, J., Chen, Y., Shao, Z., Jian, L., 2025. Grid integration of electric vehicles within electricity and carbon markets: A comprehensive overview. *ETransportation* 25, 100435. <http://dx.doi.org/10.1016/j.etrans.2025.100435>.
- Lesjak, G.S., 2023. Quantification of Flexibility-Value of Residential Electric Vehicles through the Integration into German Balancing Markets. (Master thesis). Universitat Politècnica de Catalunya. Escola Tècnica Superior d'Enginyeria Industrial de Barcelona, Berlin, Germany, URL <https://hdl.handle.net/2117/398153>, Supervisors: Andreas Sumper, Jacob Dalton.
- Li, Z., 2024. The electrification of transport: Key drivers, challenges, and future directions. *Highlights Sci. Eng. Technol.* 121, 31–35. <http://dx.doi.org/10.54097/12kcke97>.
- Liu, Q., Li, Y., Xu, T., Qian, F., Meng, H., Yao, Y., Ruan, Y., 2024. Peak shaving potential and its economic feasibility analysis of V2B mode. *J. Build. Eng.* 90, 109271. <http://dx.doi.org/10.1016/j.jobbe.2024.109271>.
- Lunde, G.A., Damm, E.V., Gade, P.A.V., Kazempour, J., 2024. Aggregator of electric vehicles bidding in nordic FCR-d markets: A chance-constrained program. URL <http://arxiv.org/pdf/2404.12818v5>.
- Magdy, G., Ali, H., Xu, D., 2021. A new synthetic inertia system based on electric vehicles to support the frequency stability of low-inertia modern power grids. *J. Clean. Prod.* 297, 126595. <http://dx.doi.org/10.1016/j.jclepro.2021.126595>.
- Manzoli, J.A., Trovão, J.P.F., Hengeler Antunes, C., 2022. Electric bus coordinated charging strategy considering V2G and battery degradation. *Energy* 254, 124252. <http://dx.doi.org/10.1016/j.energy.2022.124252>.
- Meiers, J., Frey, G., 2024. A case study of the use of smart EV charging for peak shaving in local area grids. *Energies* 17 (1), 47. <http://dx.doi.org/10.3390/en17010047>.
- Menchaca Santos, D., Thüne, P., Zepter, J.M., Marinelli, M., 2025. Business cases for degradation-aware bidirectional charging of residential users and heavy-duty vehicle fleets. *ETransportation* 23, 100389. <http://dx.doi.org/10.1016/j.etrans.2024.100389>.
- Merten, M., Rücker, F., Schoeneberger, I., Sauer, D.U., 2020. Automatic frequency restoration reserve market prediction: Methodology and comparison of various approaches. *Appl. Energy* 268, 114978. <http://dx.doi.org/10.1016/j.apenergy.2020.114978>.
- Mladenov, V., Chobanov, V., Georgiev, A., 2021. Impact of renewable energy sources on power system flexibility requirements. *Energies* 14 (10), 2813. <http://dx.doi.org/10.3390/en14102813>.
- Möbius, T., Riepin, I., Müsgens, F., van der Weijde, A.H., 2023. Risk aversion and flexibility options in electricity markets. *Energy Econ.* 126, 106767. <http://dx.doi.org/10.1016/j.eneco.2023.106767>.
- Naharudinsyah, I., Limmer, S., 2018. Optimal charging of electric vehicles with trading on the intraday electricity market. *Energies* 11 (6), 1416. <http://dx.doi.org/10.3390/en11061416>.
- Netztransparenz.de, 2025. Netztransparenz.de - activated balancing capacity. URL <https://www.netztransparenz.de/en/Balancing-Capacity/Balancing-Capacity-data/Activated-Balancing-Capacity>.
- NODES, 2025. FinFlex – TSO-DSO congestion management market. URL <https://nodesmarket.com/finflex/>.
- Ohno, Y., Inagaki, S., Yamaguchi, T., Ito, A., Suzuki, T., 2024. Charging control method for HEMS considering multiple vehicle usage scenarios by using EV. In: 2024 SICE Festival with Annual Conference (SICE FES). pp. 1006–1011.
- Ostermann, A., Engwerth, V., Sommer, K., 2022. Analysis of the peak shaving potential of bidirectionally chargeable electric vehicles in a field trial. *IET Conf. Proc.* 2022 (19), 26–32. <http://dx.doi.org/10.1049/icp.2022.2712>.
- Ostermann, J., Koetter, F., 2016. Energy-management-as-a-service: Mobility aware energy management for a shared electric vehicle fleet. In: Proceedings of the 5th International Conference on Smart Cities and Green ICT Systems. SCITEPRESS - Science and Technology Publications, pp. 340–350. <http://dx.doi.org/10.5220/0005814803400350>.
- Papadimitriou, C., Schulze, J.C., Mitsos, A., 2024. Representative electricity price profiles for European day-ahead and intraday spot markets. URL <http://arxiv.org/pdf/2405.14403v1>.
- Paredes, Á., Aguado, J.A., 2024. Revenue stacking of BESSs in wholesale and aFRR markets with delivery guarantees. *Electr. Power Syst. Res.* 234, 110633. <http://dx.doi.org/10.1016/j.epsr.2024.110633>.
- Park, S.-W., Cho, K.-S., Hoeft, G., Son, S.-Y., 2022. Electric vehicle charging management using location-based incentives for reducing renewable energy curtailment considering the distribution system. *Appl. Energy* 305, 117680. <http://dx.doi.org/10.1016/j.apenergy.2021.117680>.
- Pertl, M., Carducci, F., Tabone, M., Marinelli, M., Kiliccote, S., Kara, E.C., 2019. An equivalent time-variant storage model to harness EV flexibility: Forecast and aggregation. *IEEE Trans. Ind. Inform.* 15 (4), 1899–1910. <http://dx.doi.org/10.1109/TII.2018.2865433>.
- Piffaretti, M., Meier, R., Barth, P., Limacher, C., Stadie, M., Schopfer, S., Kramis, M., Chako, A., Erne, D., Flury, R., Stephan, A., Moser, R., Kreusel, J., Roschewitz, A., 2024. V2X suisse: Final procet report.
- Prakash, K., Ali, M., Siddique, M., Karmaker, A.K., Macana, C.A., Dong, D., Pota, H.R., 2022. Bi-level planning and scheduling of electric vehicle charging stations for peak shaving and congestion management in low voltage distribution networks. *Comput. Electr. Eng.* 102, 108235. <http://dx.doi.org/10.1016/j.compeleceng.2022.108235>.
- Regelleistung.net, 2025. Regelleistung.net - datencenter. URL <https://www.regelleistung.net/apps/datacenter/tenders/>.
- Rosner, P., Dietermann, B., Brödel, M., Paper, A., Lienkamp, M., 2025. REVOL-e-TION: A flexible and scalable investment optimization toolbox for local energy systems incorporating electric vehicle fleets. *SoftwareX* 31, 102178. <http://dx.doi.org/10.1016/j.softx.2025.102178>.
- RTE (Réseau de Transport d'Electricité), 2025. Flexibilities : provide your services for network congestion management. URL <https://www.services-rte.com/en/learn-more-about-our-services/flexibilities-provide-your-services-for-network-congestion-management.html>.
- Rücker, F., Merten, M., Gong, J., Villafañila-Robles, R., Schoeneberger, I., Sauer, D.U., 2020. Evaluation of the effects of smart charging strategies and frequency restoration reserves market participation of an electric vehicle. *Energies* 13 (12), 3112. <http://dx.doi.org/10.3390/en13123112>.
- Saadatmandi, S., Chicco, G., Favenza, A., Mozzato, A., Giordano, F., Arnone, M., 2024. Smart electric vehicle charging for reducing photovoltaic energy curtailment. *Electr. Power Syst. Res.* 230, 110181. <http://dx.doi.org/10.1016/j.epsr.2024.110181>.
- Sagaria, S., van der Kam, M., Boström, T., 2025. Vehicle-to-grid impact on battery degradation and estimation of V2g economic compensation. *Appl. Energy* 377, 124546. <http://dx.doi.org/10.1016/j.apenergy.2024.124546>.
- Seo, M., Jin, Y., Kim, M., Son, H., Han, S., 2024. Strategies for electric vehicle bidding in the german frequency containment and restoration reserves market. *Electr. Power Syst. Res.* 228, 110040. <http://dx.doi.org/10.1016/j.epsr.2023.110040>.
- Signer, T., Unger, F., Ruppert, M., Fichtner, W., 2023. Economic potential of V2G in electricity markets – a systematic literature review. In: 2023 IEEE Vehicle Power and Propulsion Conference. VPPC, pp. 1–6. <http://dx.doi.org/10.1109/VPPC60535.2023.10403286>.
- Sinsel, S.R., Riemke, R.L., Hoffmann, V.H., 2020. Challenges and solution technologies for the integration of variable renewable energy sources—a review. *Renew. Energy* 145, 2271–2285. <http://dx.doi.org/10.1016/j.renene.2019.06.147>.
- Tepe, B., Figgenger, J., Englberger, S., Sauer, D.U., Jossen, A., Hesse, H., 2022. Optimal pool composition of commercial electric vehicles in V2G fleet operation of various electricity markets. *Appl. Energy* 308, 118351. <http://dx.doi.org/10.1016/j.apenergy.2021.118351>.
- Teske, F., Fehrl, A., Praß, J., Franke, J., 2021. Adapted pricing scheme for the integration of vehicle-to-grid into the energy system. In: 2021 11th International Electric Drives Production Conference. EDPC, pp. 1–7. <http://dx.doi.org/10.1109/EDPC53547.2021.9684223>.
- Thomaßen, G., Redl, C., Bruckner, T., 2022. Will the energy-only market collapse? On market dynamics in low-carbon electricity systems. *Renew. Sustain. Energy Rev.* 164, 112594. <http://dx.doi.org/10.1016/j.rser.2022.112594>.
- Thompson, A.W., Perez, Y., 2020. Vehicle-to-everything (V2X) energy services, value streams, and regulatory policy implications. *Energy Policy* 137, 111136. <http://dx.doi.org/10.1016/j.enpol.2019.111136>.
- van Krieking, G., de Cauwer, C., Sapountzoglou, N., Coosemans, T., Messagie, M., 2021. Peak shaving and cost minimization using model predictive control for uni- and bi-directional charging of electric vehicles. *Energy Rep.* 7, 8760–8771. <http://dx.doi.org/10.1016/j.egy.2021.11.207>.
- Visser, L.R., Kootte, M.E., Ferreira, A.C., Sicurani, O., Pauwels, E.J., Vuik, C., van Sark, W., Alskaf, T.A., 2022. An operational bidding framework for aggregated electric vehicles on the electricity spot market. *Appl. Energy* 308, 118280. <http://dx.doi.org/10.1016/j.apenergy.2021.118280>.
- Vollmuth, P., Wohlschlager, D., Wasmeier, L., Kern, T., 2024. Prospects of electric vehicle V2G multi-use: Profitability and ghg emissions for use case combinations of smart and bidirectional charging today and 2030. *Appl. Energy* 371, 123679. <http://dx.doi.org/10.1016/j.apenergy.2024.123679>.
- Wang, Z., Raj, P.J., Babu, B.R., Ramaiah, G.B., 2024. Innovative EV charging optimization: Fuzzy DSS algorithm unveils benefits. *Int. J. Fuzzy Syst.* <http://dx.doi.org/10.1007/s40815-024-01881-2>.
- Wei, M., McMillan, C.A., de la Rue Can, S., 2019. Electrification of industry: Potential, challenges and outlook. *Curr. Sustain. Renew. Energy Rep.* 6 (4), 140–148. <http://dx.doi.org/10.1007/s40518-019-00136-1>.
- Weiß, A., Müller, M., Franz, S., 2021. Spitzenlastkappung durch uni- und bidirektionales laden von elektrofahrzeugen und analyse der resultierenden netzbelastung in verteilnetzen. *Forsch. Im Ingenieurwesen* 85 (2), 469–476. <http://dx.doi.org/10.1007/s10010-020-00424-z>.

- Wen, Y., Chu, Z., Srivastava, A., Teng, F., Chen, B., 2025. Resilient microgrid scheduling with synthetic inertia from electric vehicles within a network of charging stations. *IEEE Trans. Control Syst. Technol.* 33 (2), 775–787. <http://dx.doi.org/10.1109/TCST.2024.3512432>.
- Will, C., Ocker, F., 2024. Flexibility potential of smart charging electric trucks and buses. *World Electr. Veh. J.* 15 (2), 56. <http://dx.doi.org/10.3390/wevj15020056>.
- Xu, Z., Hu, Z., Song, Y., Wang, J., 2017. Risk-averse optimal bidding strategy for demand-side resource aggregators in day-ahead electricity markets under uncertainty. *IEEE Trans. Smart Grid* 8 (1), 96–105. <http://dx.doi.org/10.1109/TSG.2015.2477101>.
- Yoshimura, Y., Kondo, T., Kawanishi, M., Narikiyo, T., Sato, A., 2015. Model predictive control of EV storage battery with HEMS based on particle swarm optimization. In: 2015 IEEE Innovative Smart Grid Technologies - Asia (ISGT ASIA). IEEE, pp. 1–5. <http://dx.doi.org/10.1109/ISGT-Asia.2015.7387006>.
- Yousefi, M., Hajizadeh, A., Soltani, M.N., Hredzak, B., 2021. Predictive home energy management system with photovoltaic array, heat pump, and plug-in electric vehicle. *IEEE Trans. Ind. Inform.* 17 (1), 430–440. <http://dx.doi.org/10.1109/TII.2020.2971530>.
- Yusuf, J., Hasan, A.S.M.J., Ula, S., 2021. Impacts analysis & field implementation of plug-in electric vehicles participation in demand response and critical peak pricing for commercial buildings. In: 2021 IEEE Texas Power and Energy Conference. TPEC, IEEE, pp. 1–6. <http://dx.doi.org/10.1109/TPEC51183.2021.9384937>.
- Zachmann, G., Hirth, L., Heussaff, C., Schlecht, I., Mühelnpfordt, J., Eicke, A., 2023. The Design of the European Electricity Market Current Proposals and Ways Ahead : Study Requested by the ITRE Committee Georg Zachmann, Lion Hirth, Conall Heussaff, Ingmar Schlecht, Jonathan Mühelnpfordt and Anselm Eicke ; European Parliament, Directorate-General for Internal Policies, Policy Department for Economic, Scientific and Quality of Life Policies. European Parliament, Luxembourg.
- Zheng, S., Tu, L., Chen, X., Zhang, Z., 2025. Exploring locational marginal price volatility for arbitrage opportunities with energy storage systems in ERCOT. In: 2025 IEEE Electrical Energy Storage Applications and Technologies Conference. EESAT, pp. 1–5. <http://dx.doi.org/10.1109/EESAT62935.2025.10891236>.
- Zheng, Y., Yu, H., Shao, Z., Jian, L., 2020. Day-ahead bidding strategy for electric vehicle aggregator enabling multiple agent modes in uncertain electricity markets. *Appl. Energy* 280, 115977. <http://dx.doi.org/10.1016/j.apenergy.2020.115977>.
- Zhong, J., Kang, C., Liu, K., 2010. Demand side management in China. In: IEEE PES General Meeting. IEEE, pp. 1–4. <http://dx.doi.org/10.1109/PES.2010.5589964>.
- Zhou, M., Wu, Z., Wang, J., Li, G., 2021. Forming Dispatchable Region of electric vehicle aggregation in microgrid bidding. *IEEE Trans. Ind. Inform.* 17 (7), 4755–4765. <http://dx.doi.org/10.1109/TII.2020.3020166>.